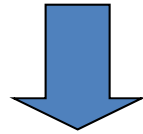


Chapter 7 Flip-Flops and Related Devices

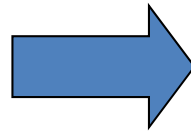
触发器以及相关设备

- **Logic circuits**

- **Combinational Circuits (组合电路)**
- **Sequential Circuits (时序电路)**



Basic Block



**Latches
Flip-Flops**

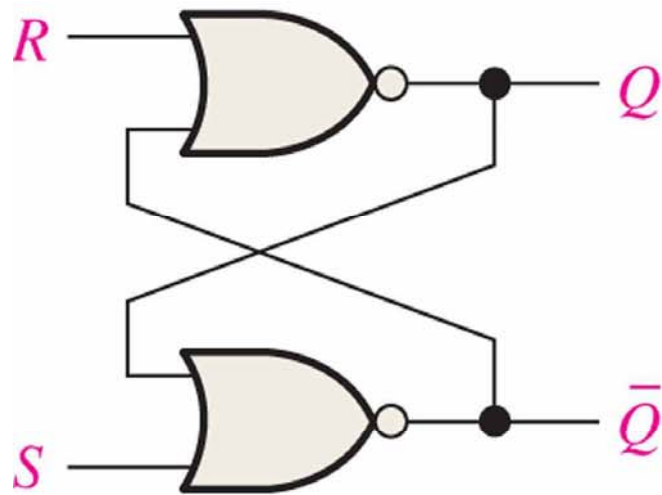
7.1 Latches (锁存器)

- The *latch* is a type of temporary device that has two stable states (*bistable*) and is normally placed in a category separate from that of *flip-flops*(触发器).
- The difference between latches and flip-flops is the method used for changing their states.

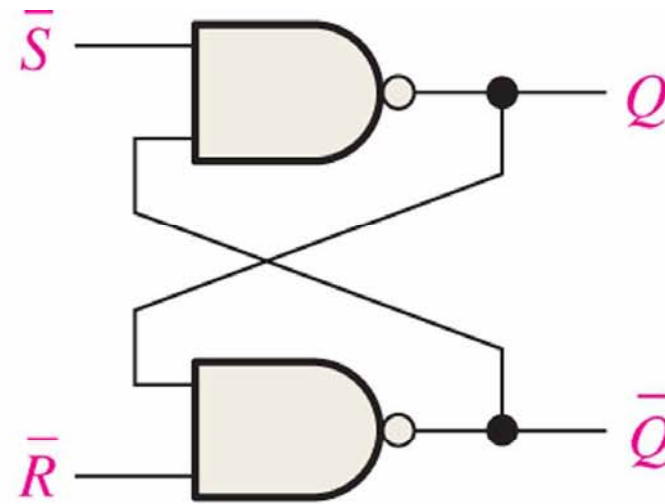
- *Latches*

- S-R (Set-Reset) Latch (SR锁存器)
- Gated S-R Latch (门控锁存器)
- Gated D (Data) Latch

7.1.1 S-R (Set-Reset) Latch

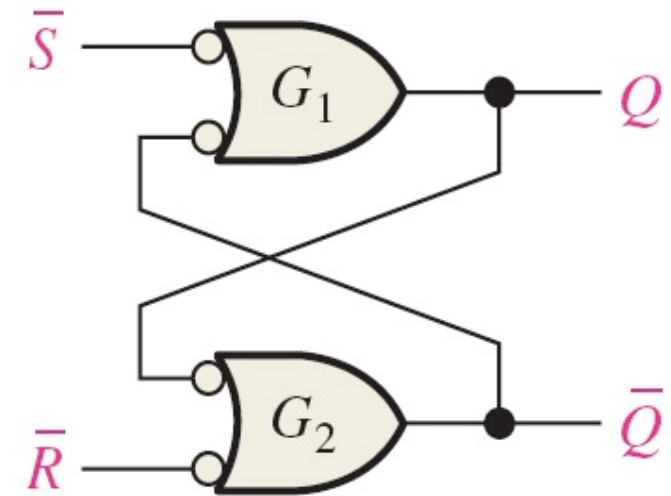
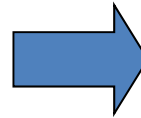
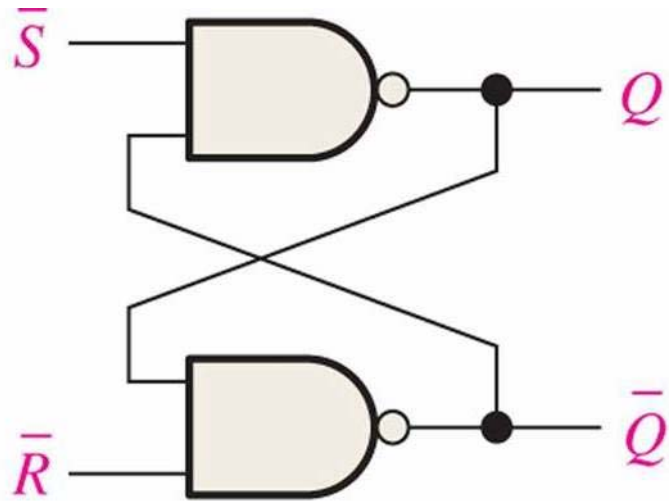


(a) Active-HIGH input S-R latch

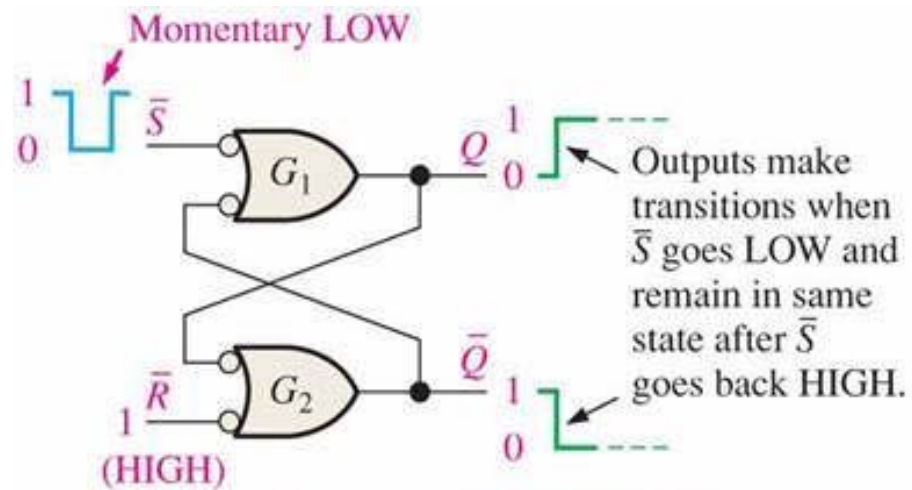


(b) Active-LOW input $\bar{S}$ - $\bar{R}$ latch

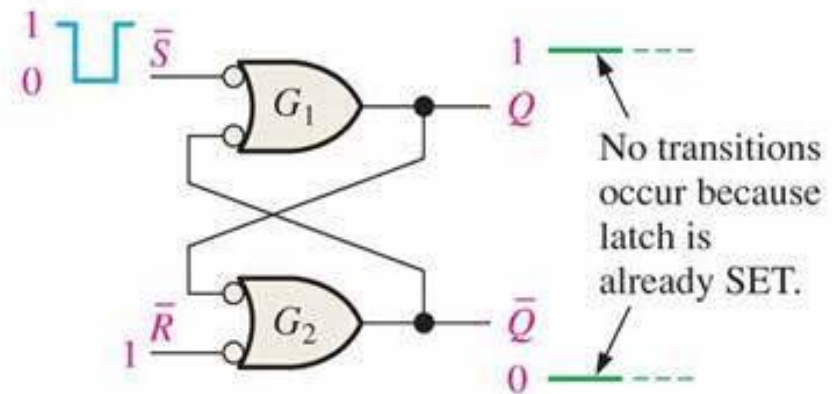
Active-LOW input S-R latch



(b) Active-LOW input $\bar{S}$ - $\bar{R}$ latch

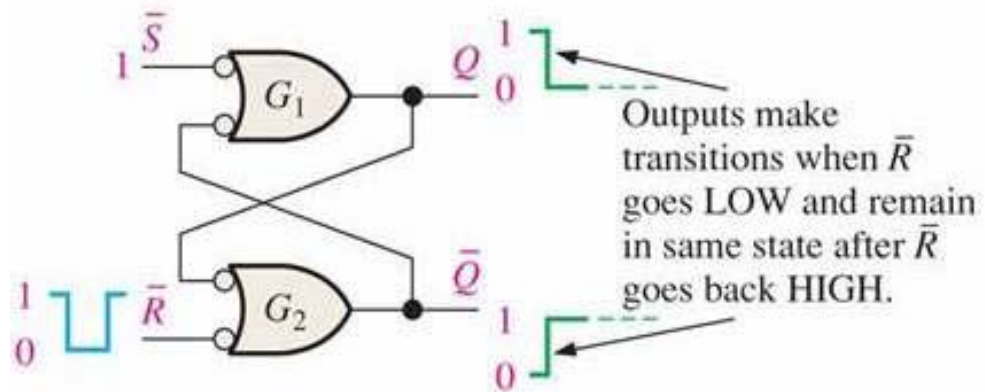


Latch starts out RESET ($Q = 0$).

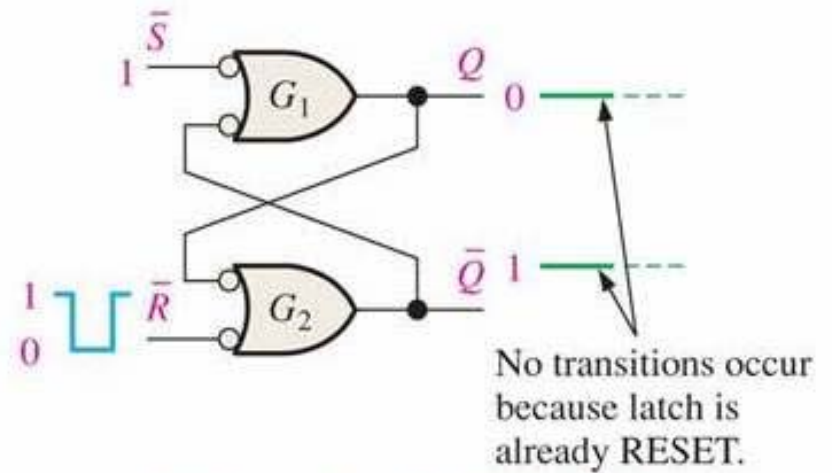


Latch starts out SET ($Q = 1$).

(a) Two possibilities for the SET operation

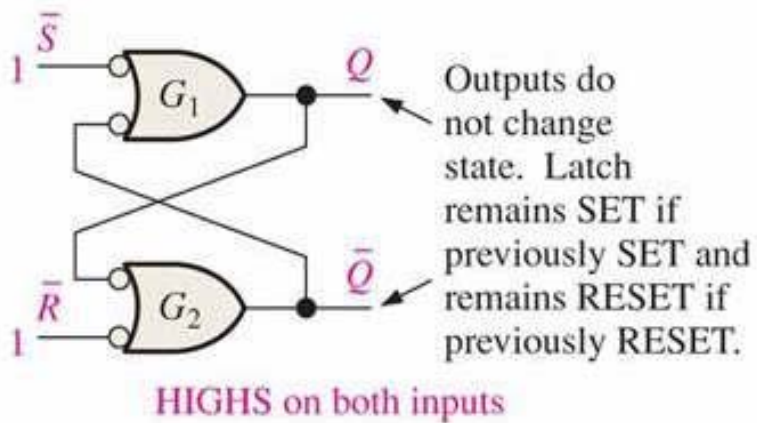


Latch starts out SET ($Q = 1$).

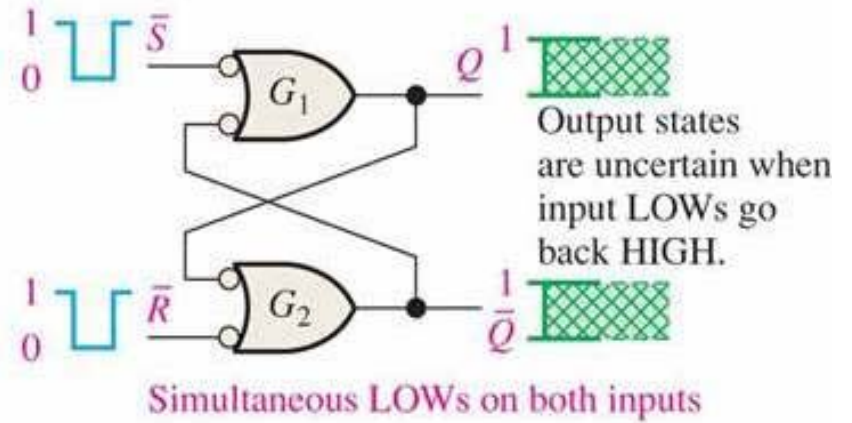


Latch starts out RESET ($Q = 0$).

(b) Two possibilities for the RESET operation



(c) No-change condition

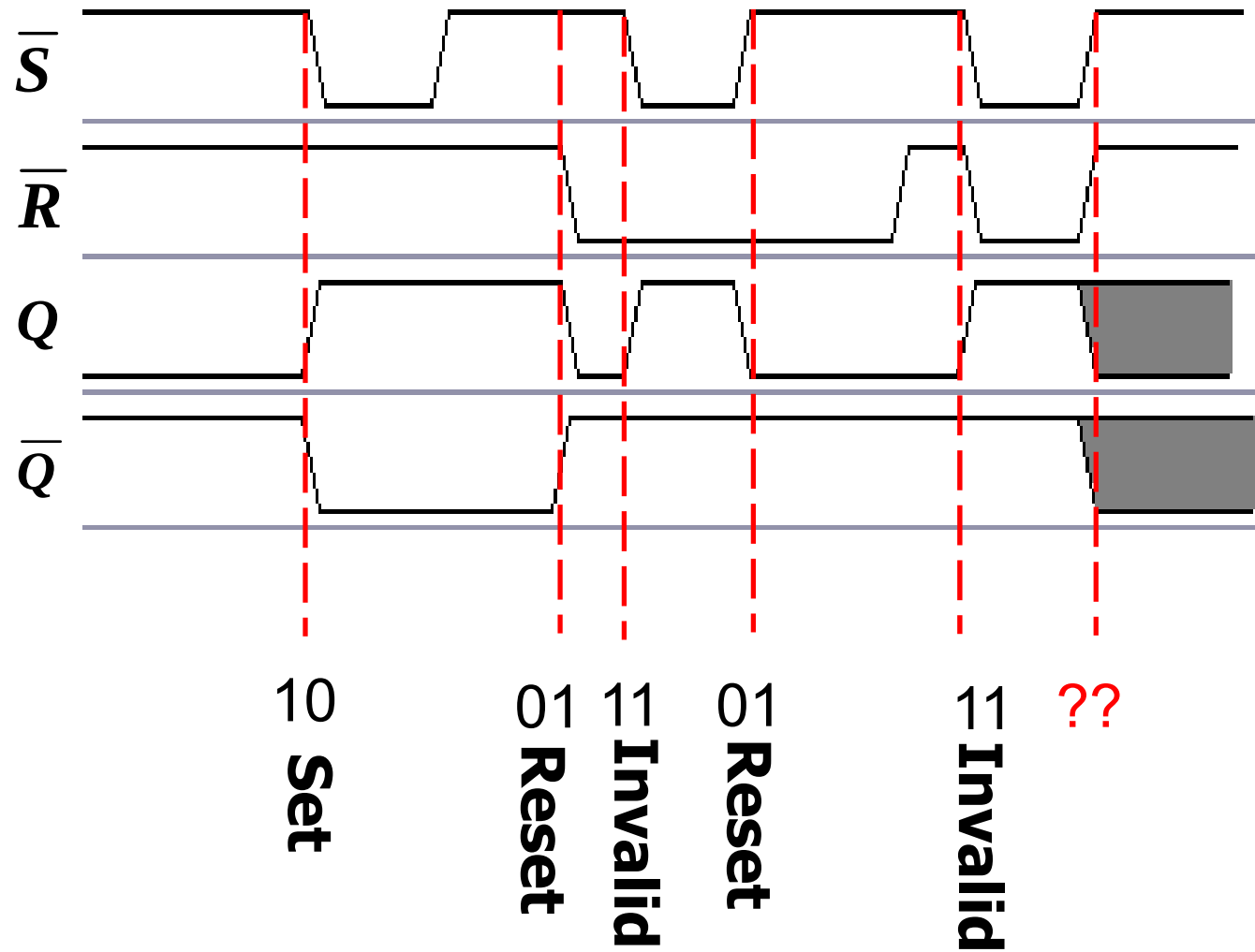


(d) Invalid condition

Truth Table (真值表)

Inputs		Outputs		Comments
$\bar{S}$	$\bar{R}$	Q	$\bar{Q}$	
1	1	NC	NC	No change. Latch remains in present state.
0	1	1	0	Latch SET.
1	0	0	1	Latch RESET.
0	0	1*	1*	Invalid condition

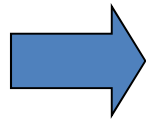
Timing diagram



Memory Function

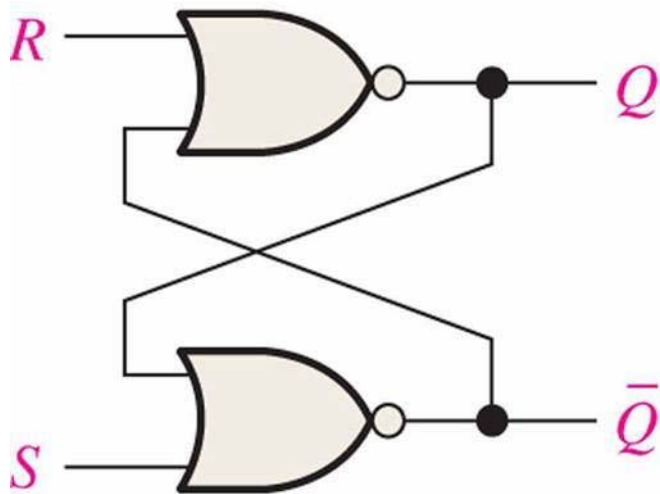
$$\begin{array}{l} S=1; \quad R=0; \left\{ \begin{array}{l} Q=0 \\ \bar{Q}=1 \end{array} \right. \quad \text{“0”, remember “0”} \\ \\ S=0; \quad R=1; \left\{ \begin{array}{l} Q=1 \\ \bar{Q}=0 \end{array} \right. \quad \text{“1”, remember “1”} \end{array}$$

Bi-stable



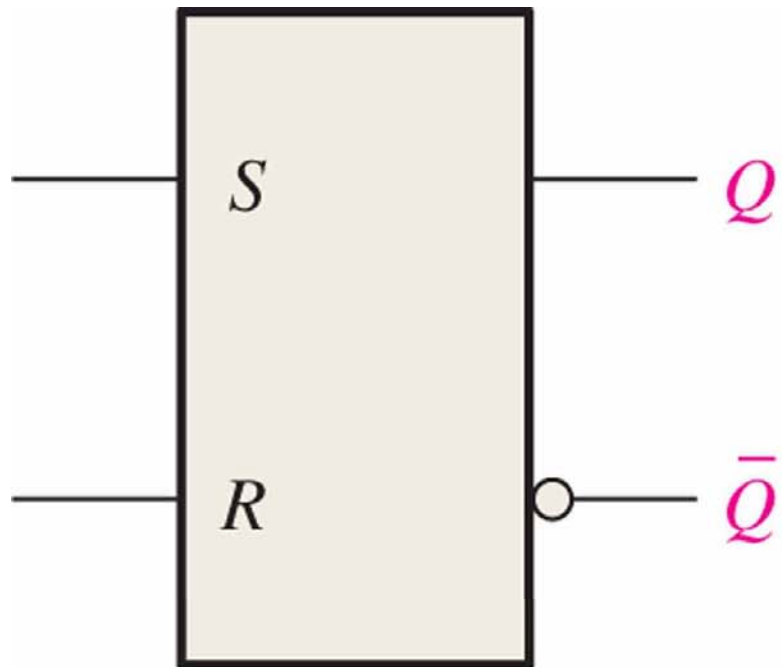
**Used to store 1 bit
binary number**

Active-HIGH input S-R latch

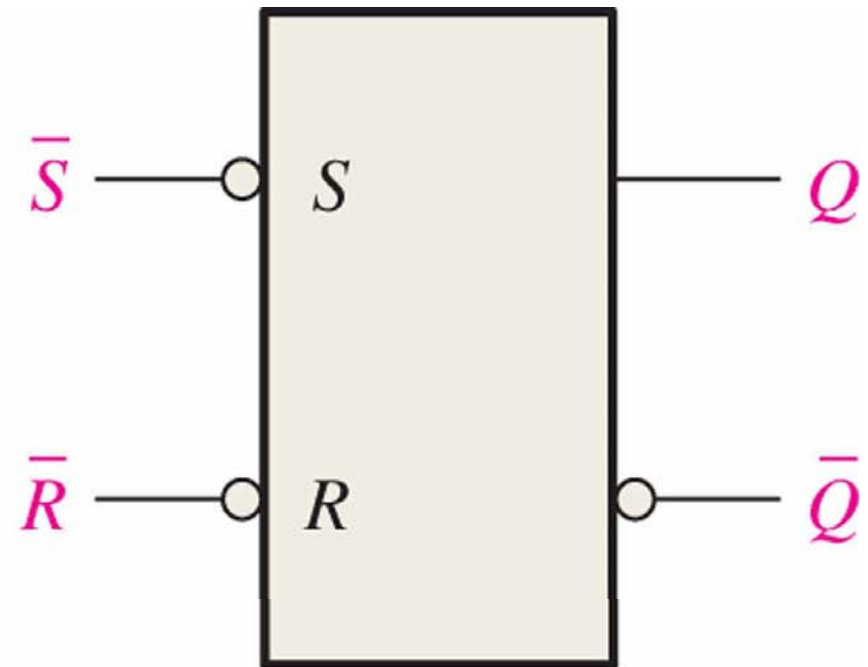


R	S	Q	$\bar{Q}$
0	0	Q_0	$\bar{Q}_0$
0	1	1	0
1	0	0	1
1	1	0^*	0^*

Logic Symbol

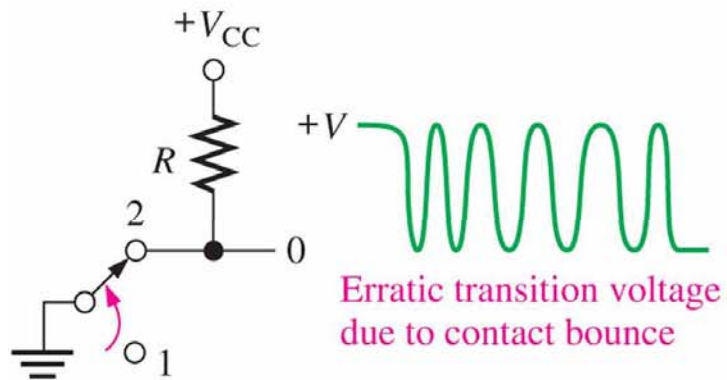


(a) Active-HIGH input
S-R latch

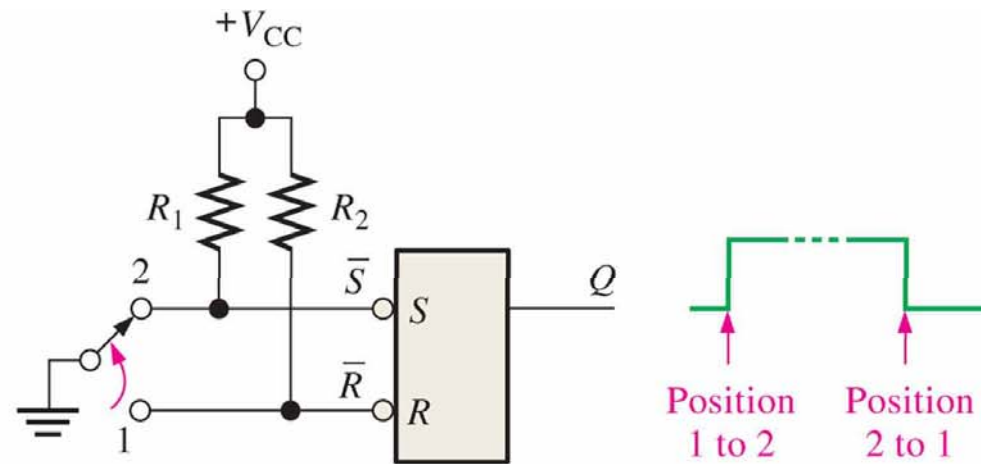


(b) Active-LOW input
 $\bar{S}$ - $\bar{R}$ latch

Application Example

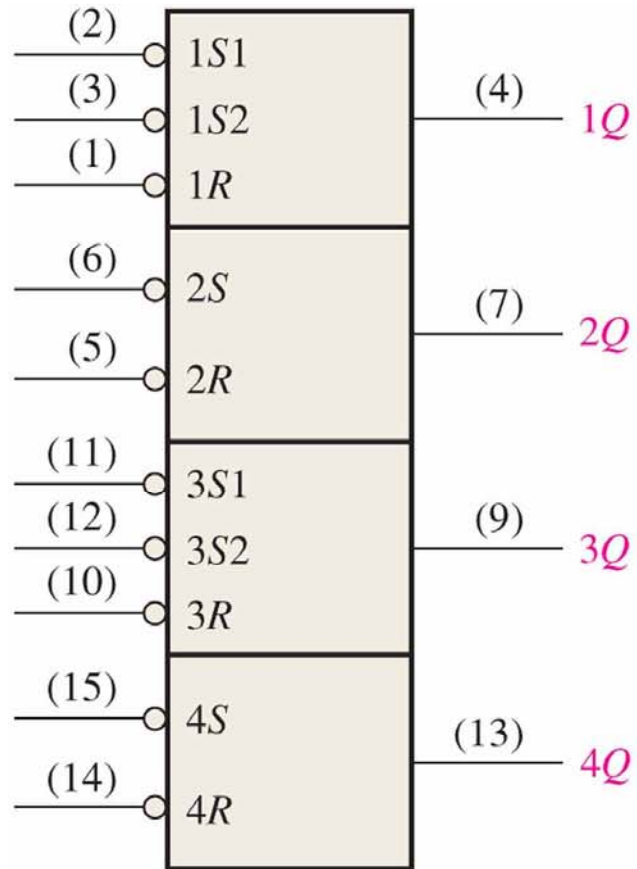


(a) Switch contact bounce

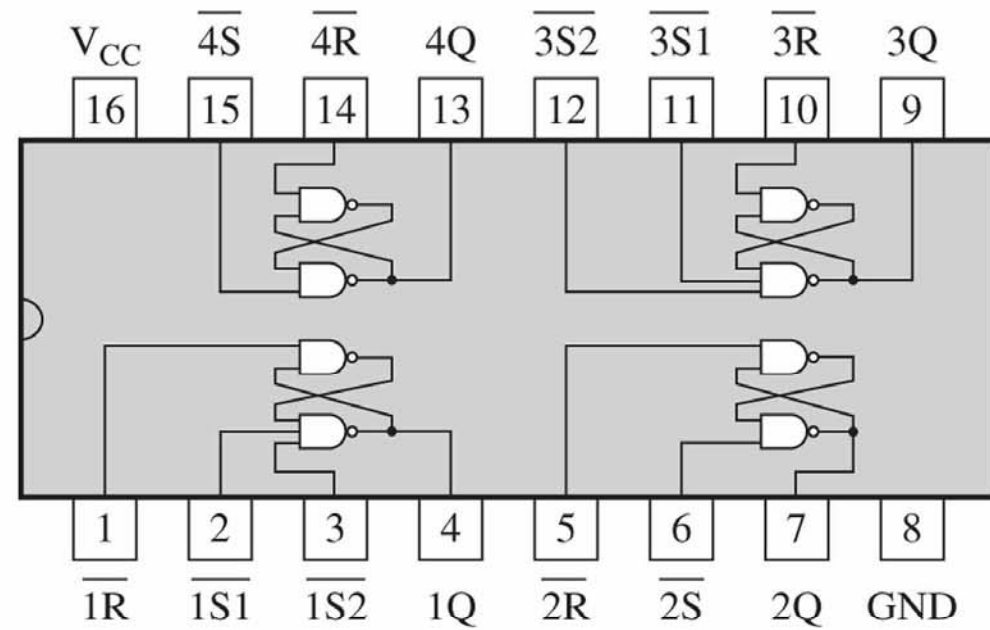


(b) Contact-bounce eliminator circuit

74LS279



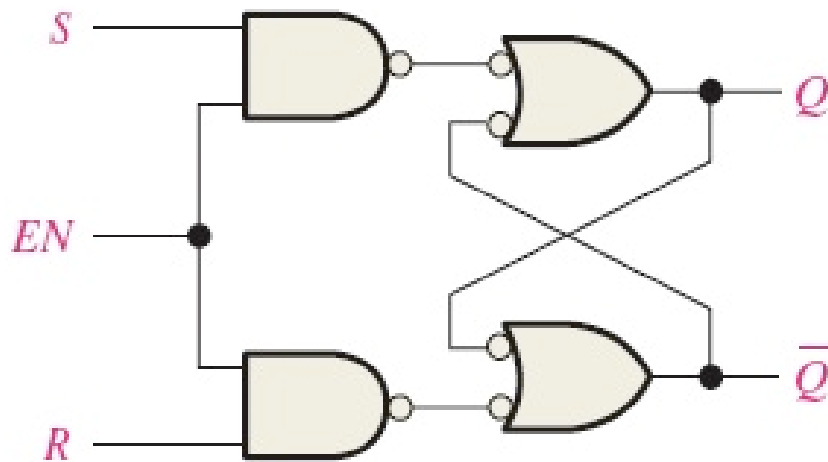
(a) Logic diagram



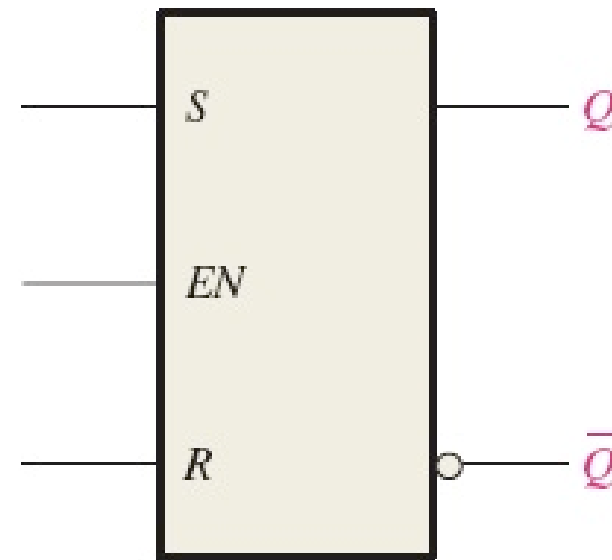
(b) Pin diagram

7.1.2 The Gated S-R Latch (门控S-R锁存器)

EN: control the time when the inputs S and R can control the output. (Level-triggered)

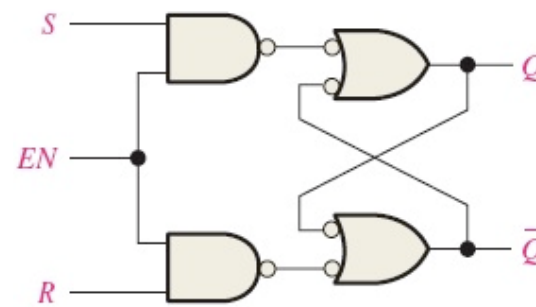
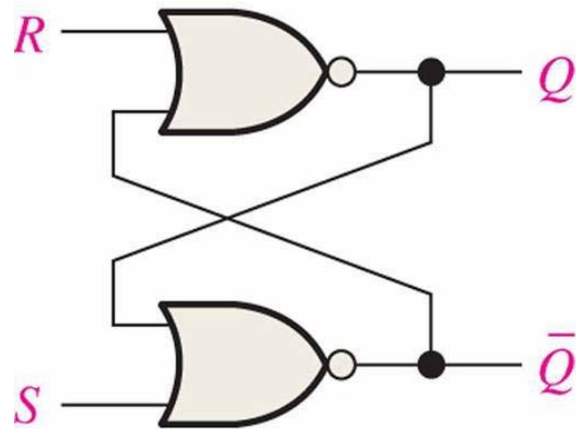


(a) Logic diagram

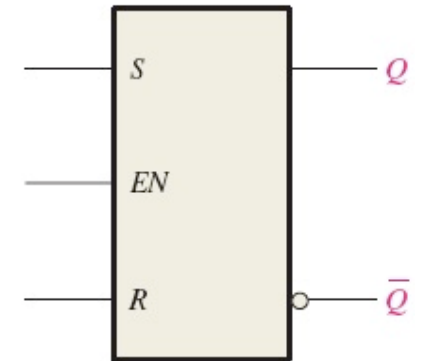


(b) Logic symbol

Comparison



(a) Logic diagram

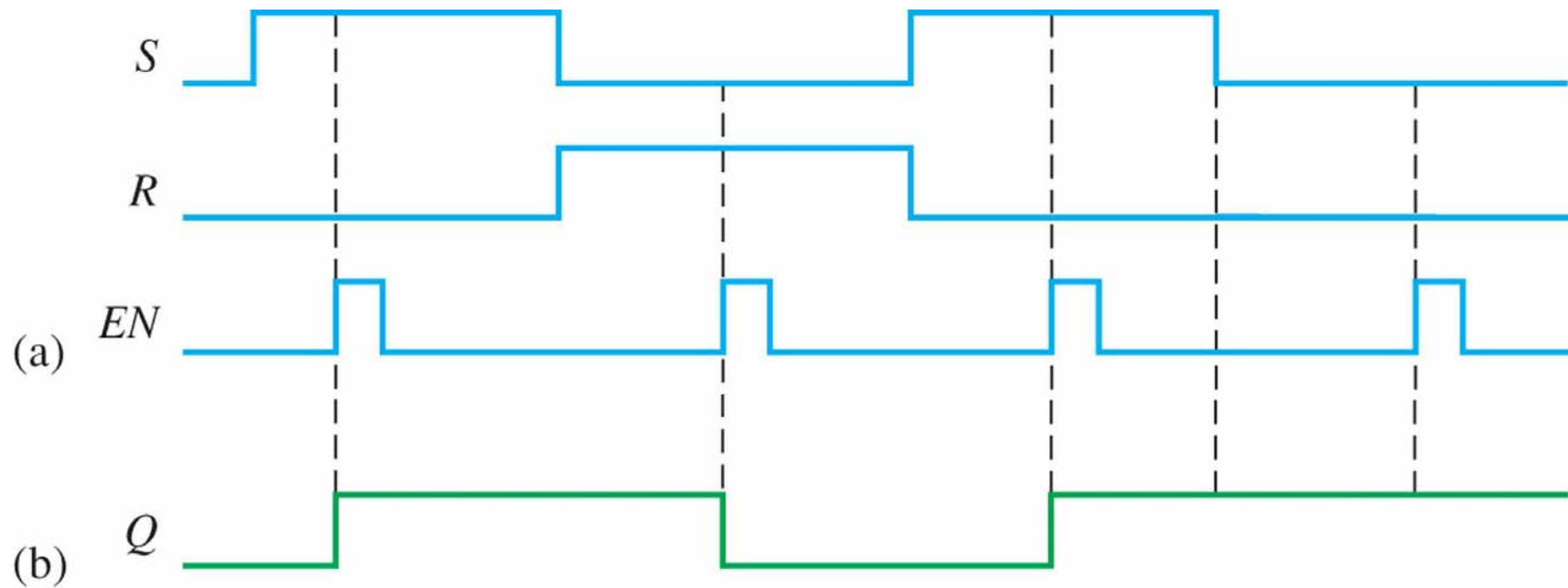


(b) Logic symbol

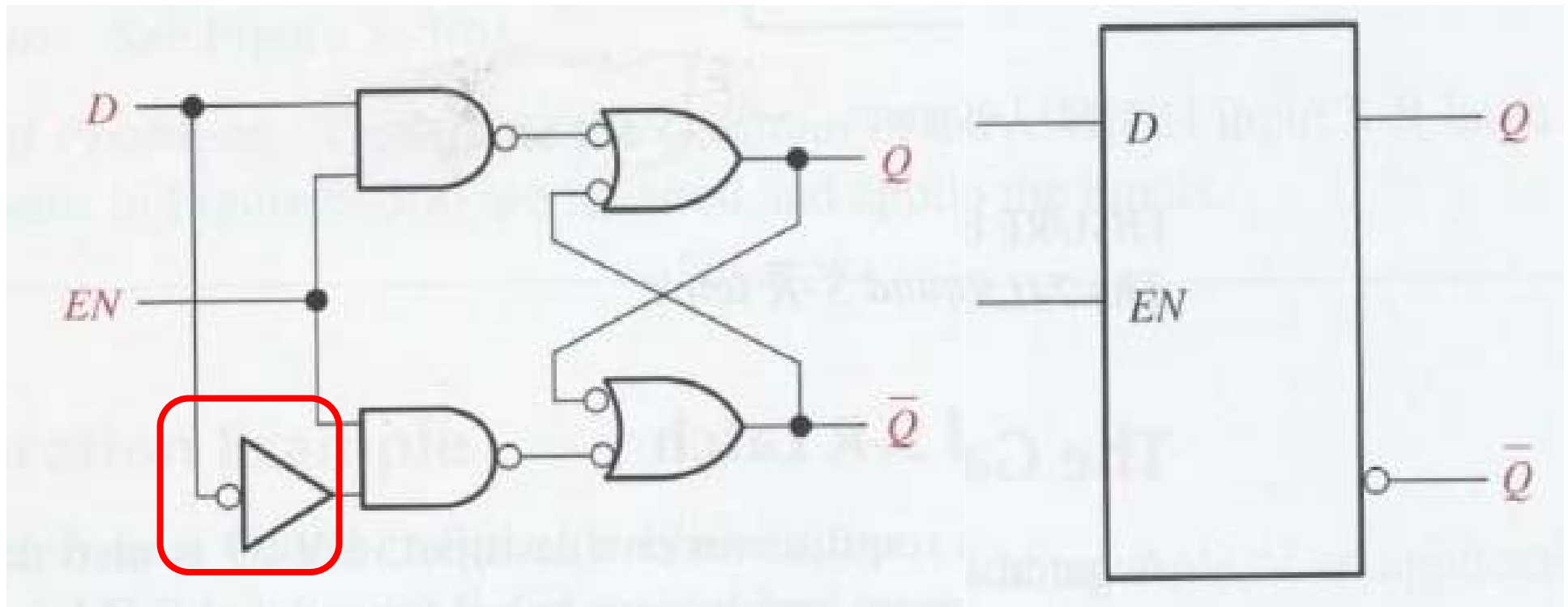
R	S	Q	$\bar{Q}$
0	0	Q_0	$\bar{Q}_0$
0	1	1	0
1	0	0	1
1	1	1^*	1^*

E	R	S	Q	$\bar{Q}$
0	X	X	Q_0	$\bar{Q}_0$
1	0	0	Q_0	$\bar{Q}_0$
1	0	1	1	0
1	1	0	0	1
1	1	1	1^*	1^*

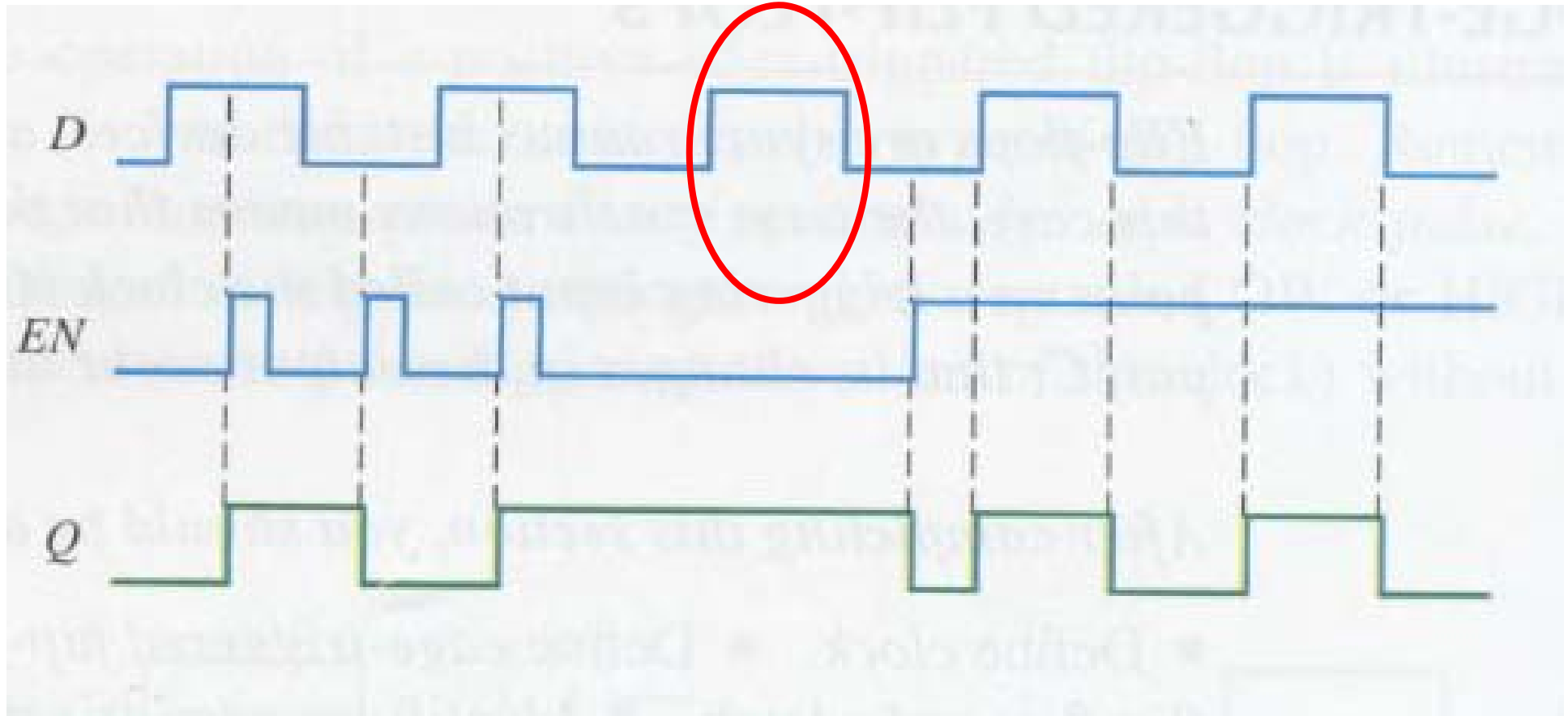
Gated S-R Latch Waveform



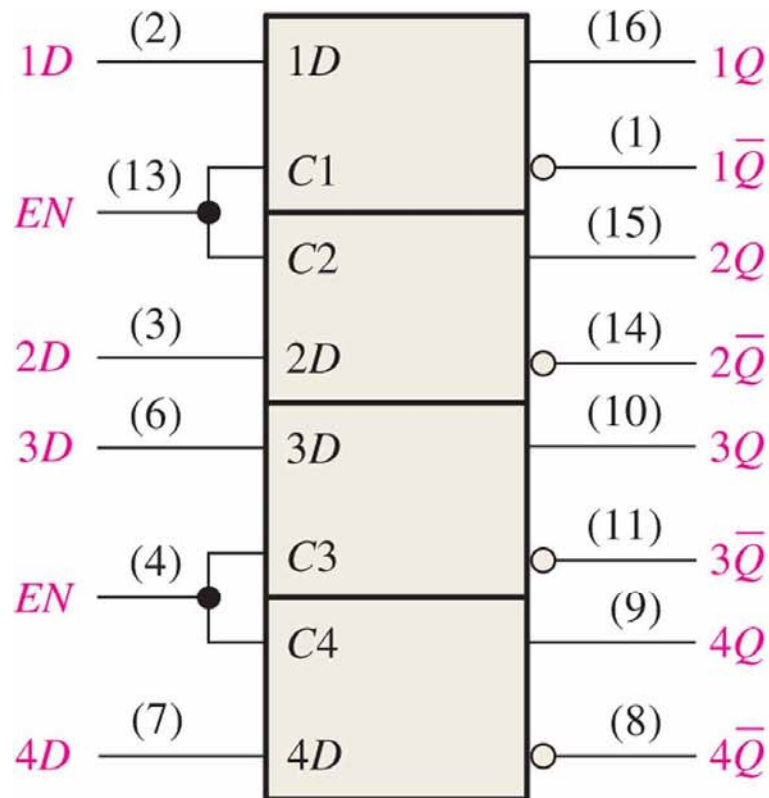
7.1.3 The Gated D Latch



Gated D Latch Waveform



74LS75



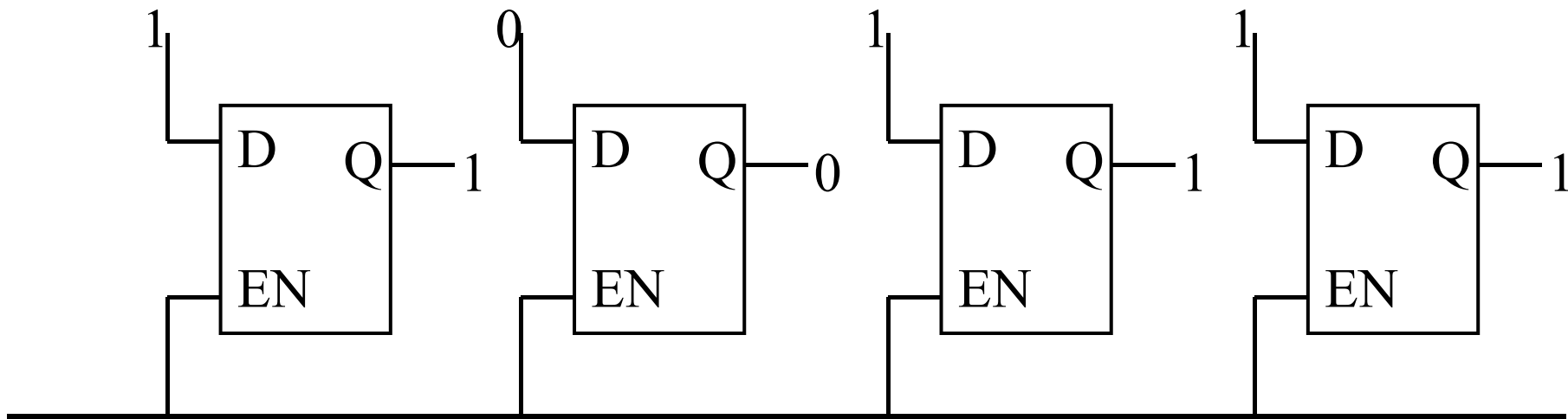
(a) Logic symbol

INPUTS		OUTPUTS		COMMENTS
D	EN	Q	$\bar{Q}$	
0	1	0	1	RESET
1	1	1	0	SET
X	0	Q_0	$\bar{Q}_0$	No change

Note: Q_0 is the prior output level before the indicated input conditions were established.

(b) Truth table (each latch)

Application example: Latches for temporary data storage



7.2 Edge-triggered Flip-Flops

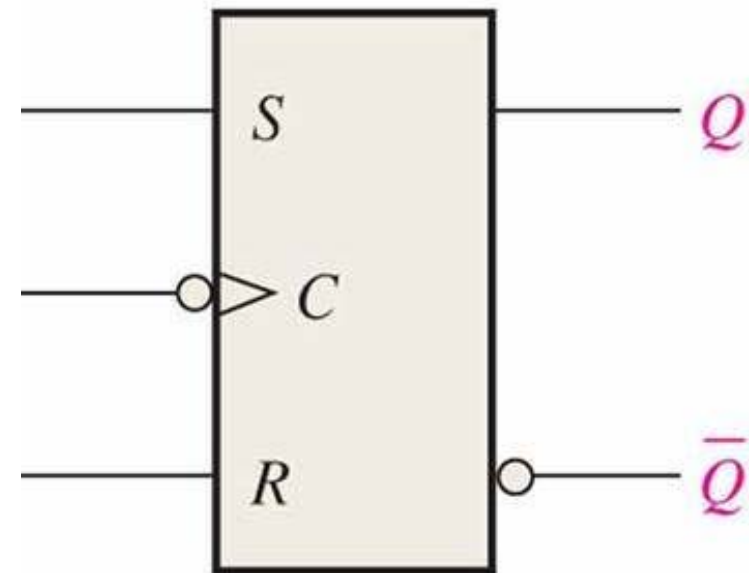
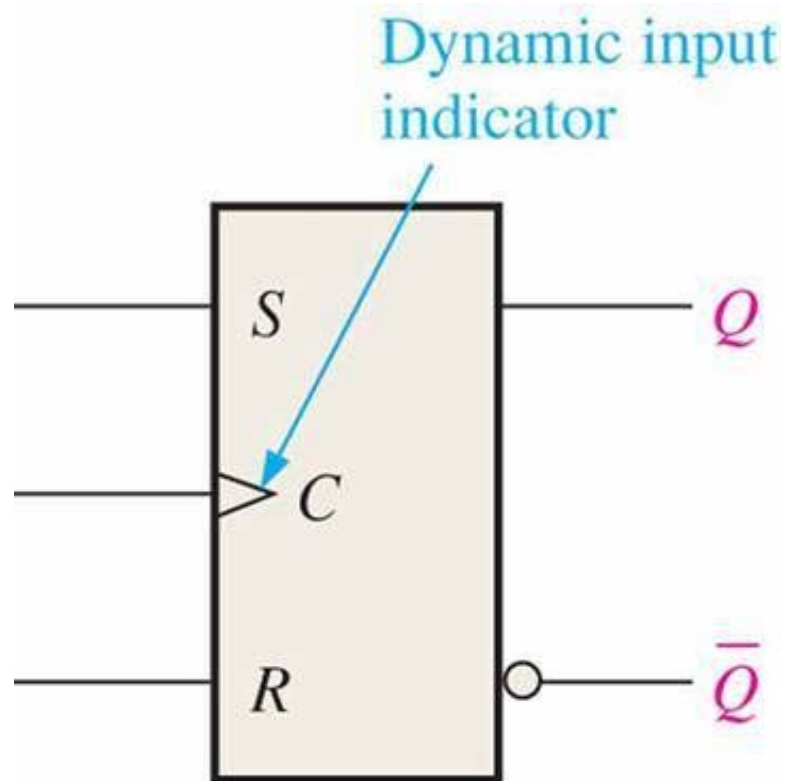
边沿触发的触发器

- Flip-flops are *synchronous bi-stable* devices. The term *synchronous* means that the output changes state only at a specified point on a *triggering input* called the *clock* (CLK) is designated as a control input C. i.e., changes in the output occur in synchronous with the clock.
- An edge-triggered flip-flop changes state either at the *positive edge* (rising edge) or at the *negative edge* (falling edge) of the clock pulse and is sensitive to its input only at this transition of the clock.

7.2.1 The edge-triggered S-R FF

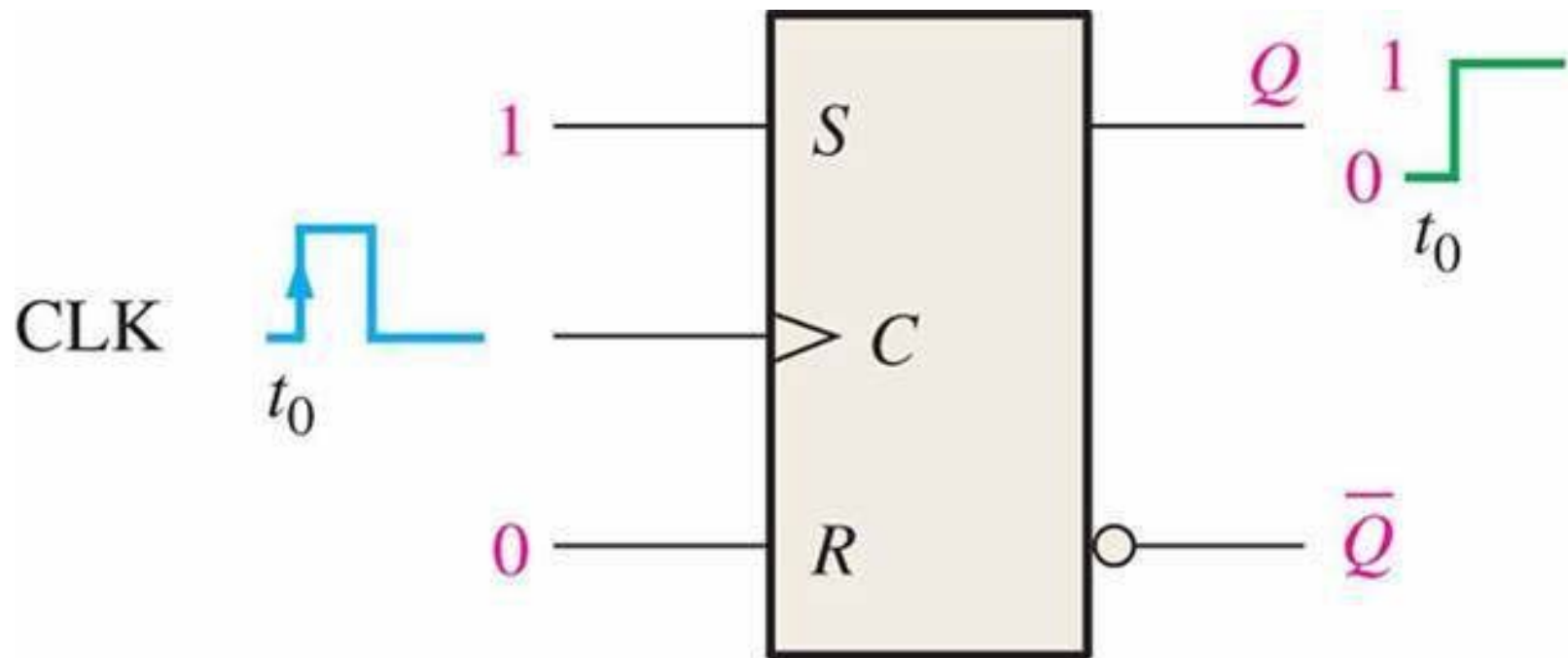
- The S and R inputs of the S-R flip-flop are called synchronous inputs because data on these inputs are transferred to the flip-flop's output only on the *triggering edge of the clock pulse*.
- When S is HIGH and R is LOW on the triggering edge of the clock pulse, Q is HIGH and the flip-flop is **SET**. When S is LOW, R is HIGH, Q is LOW and flip-flop is **RESET**. When S and R are both LOW, the output does not change from its prior state. When S and R are both HIGH, it is an invalid condition.

The edge-triggered S-R FF

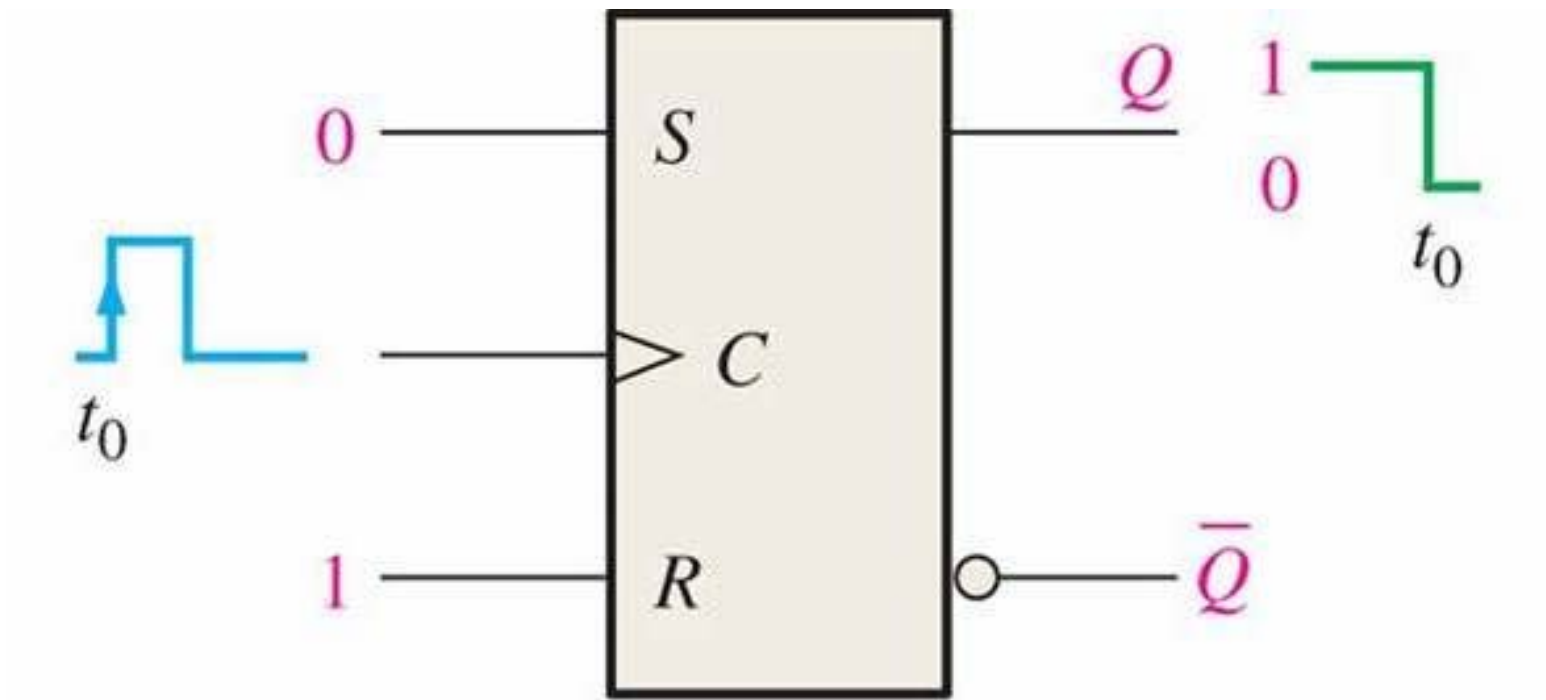


Truth Table of S-R FF

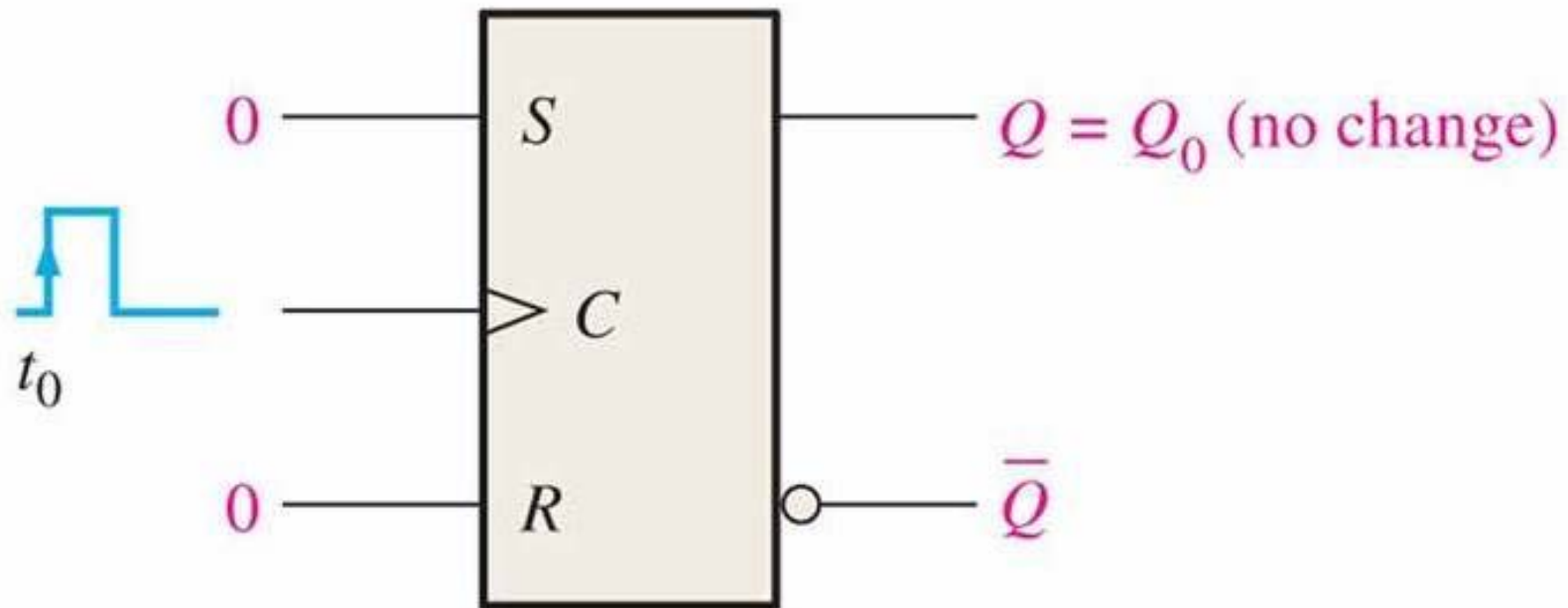
Inputs			Outputs		Comments
S	R	CLK	Q	$\bar{Q}$	
0	0	X	Q_0	$\bar{Q}_0$	No change
0	1	↑	0	1	RESET
1	0	↑	1	0	SET
1	1	↑	?	?	Invalid



(a) $S = 1, R = 0$ flip-flop SETS on positive clock edge. (If already SET, it remains SET.)

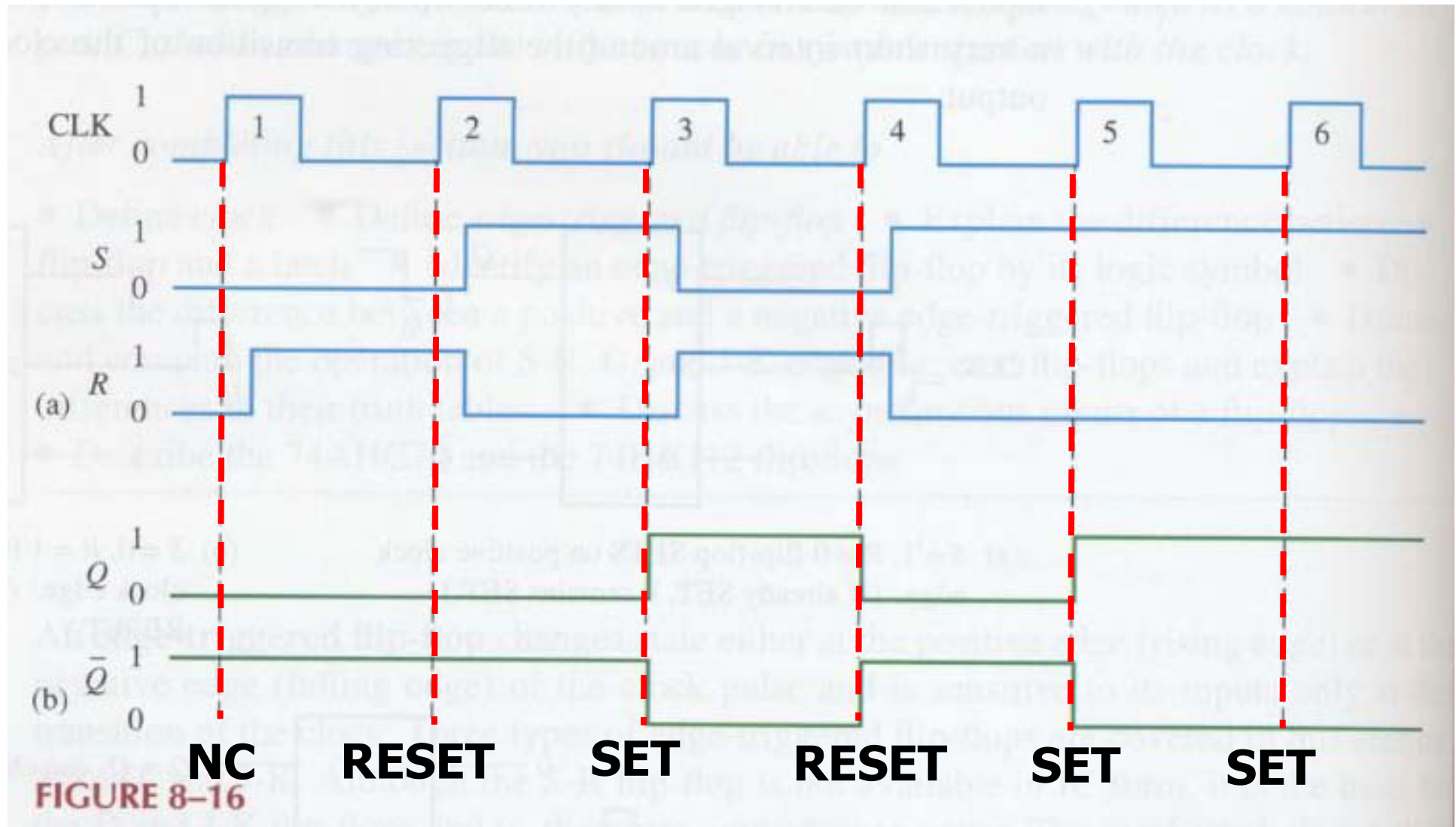


(b) $S = 0$, $R = 1$ flip-flop RESETS on positive clock edge. (If already RESET, it remains RESET.)

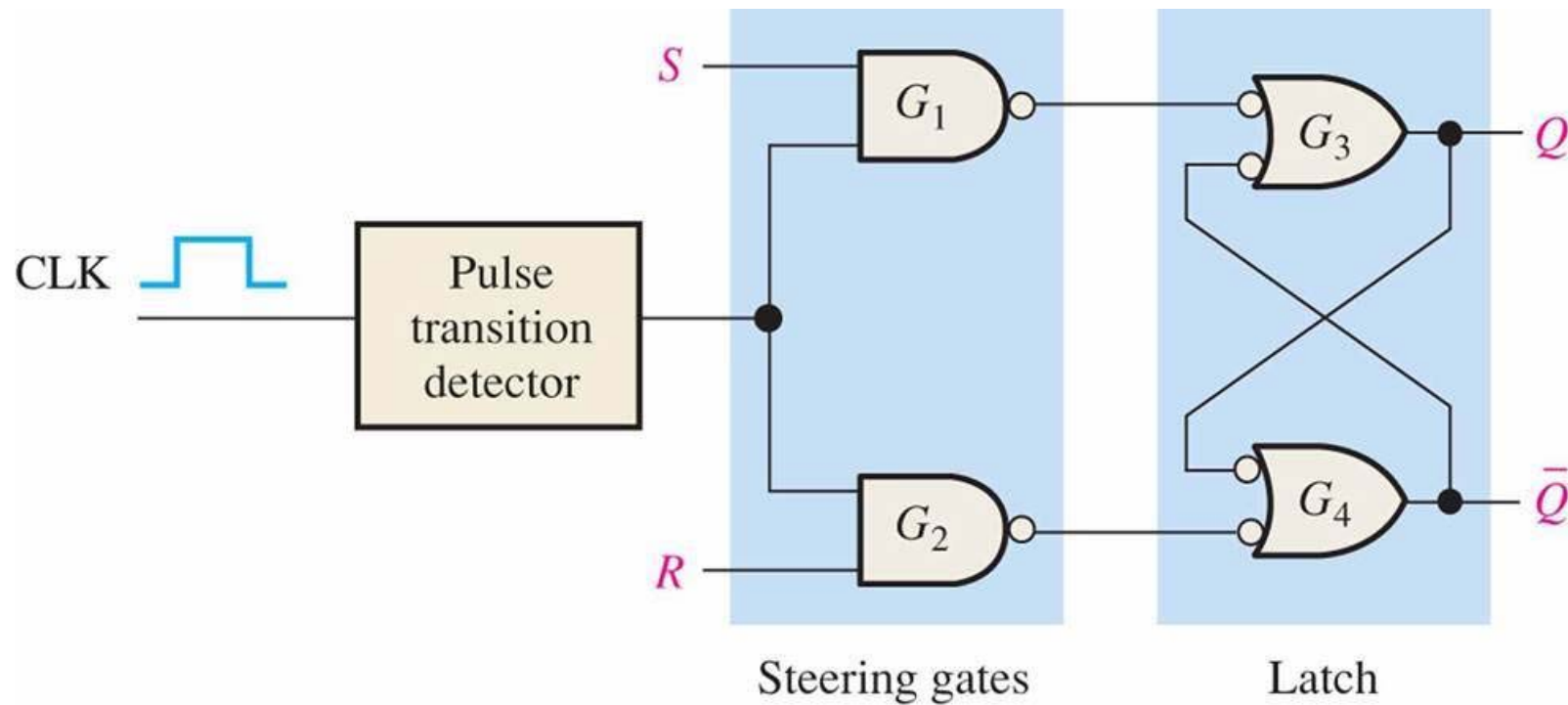


(c) $S = 0$, $R = 0$ flip-flop does not change. (If SET, it remains SET; if RESET, it remains RESET.)

Example



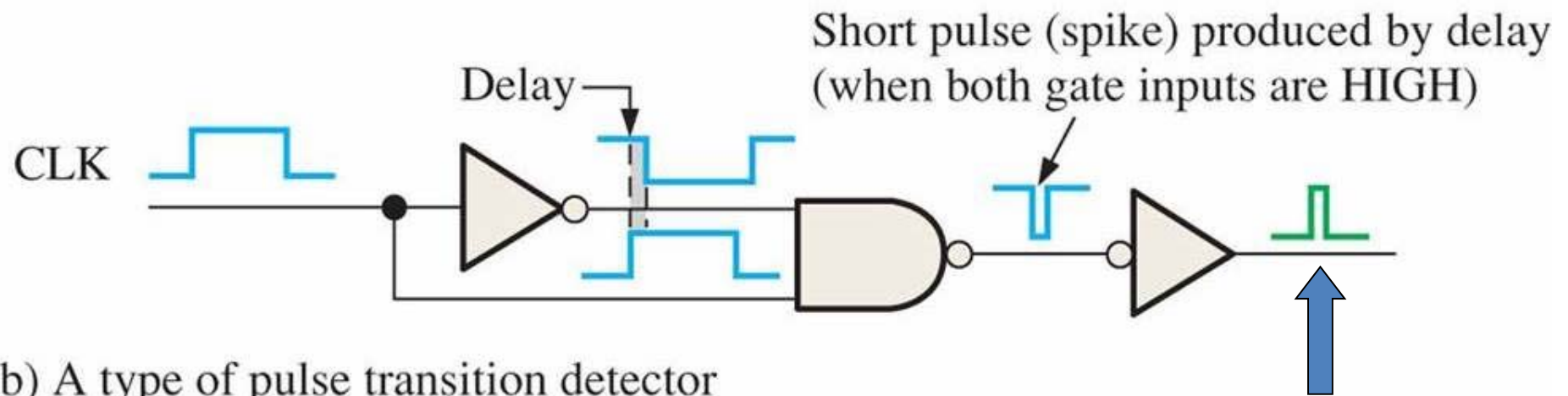
A method of Edge-triggering



(a) A simplified logic diagram for a positive edge-triggered S-R flip-flop

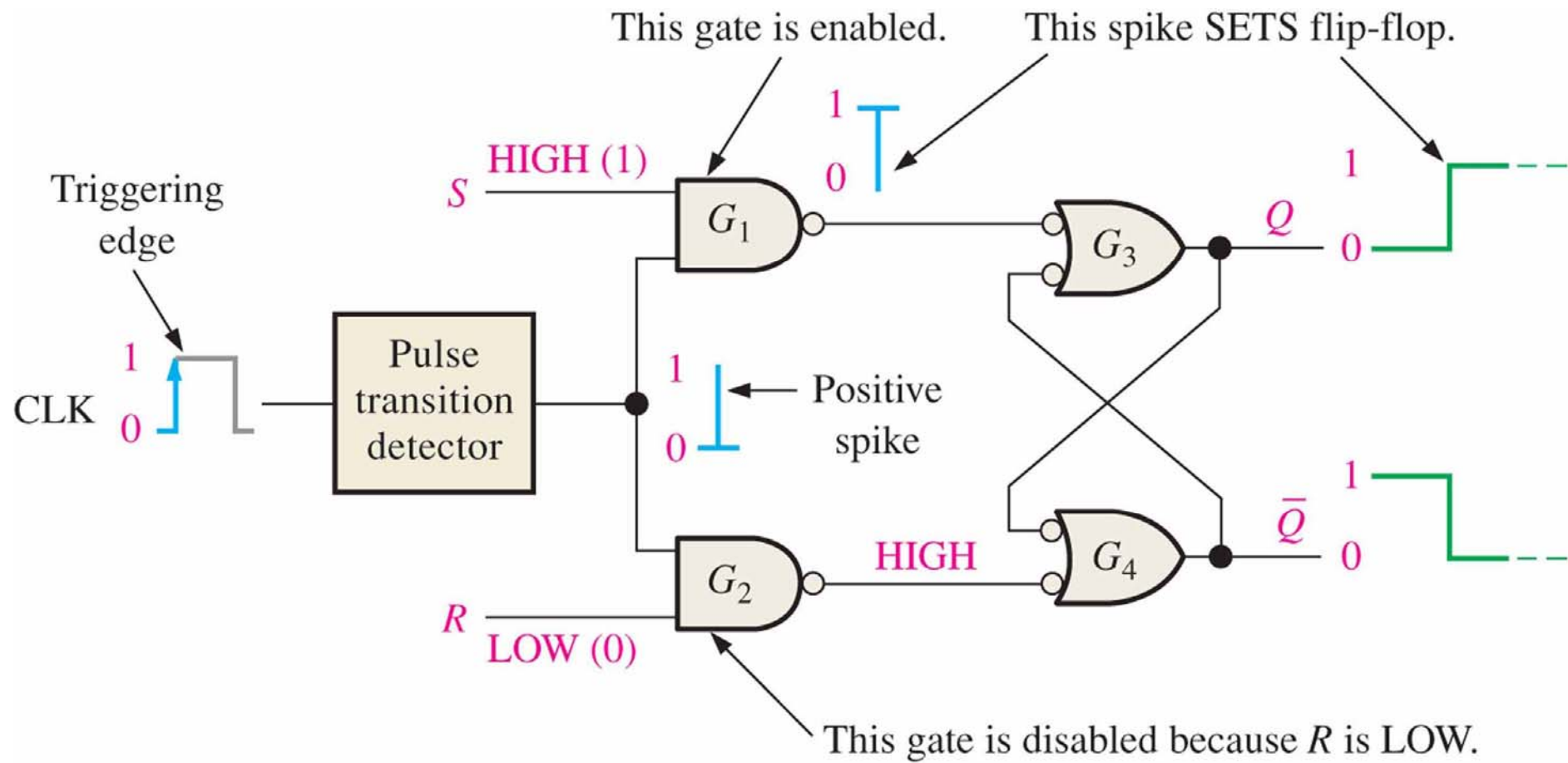
Pulse transition detector

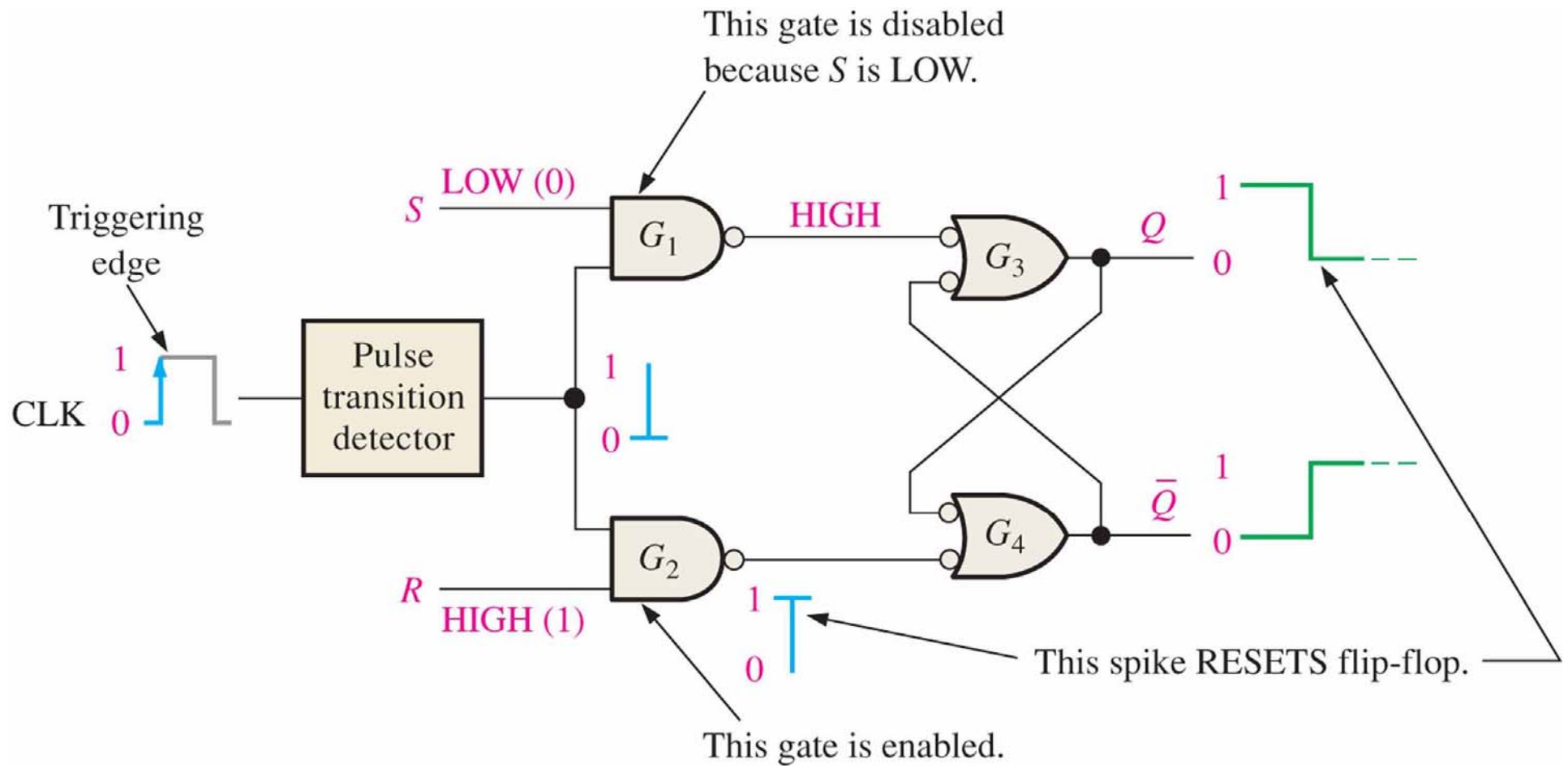
脉冲变换检测器



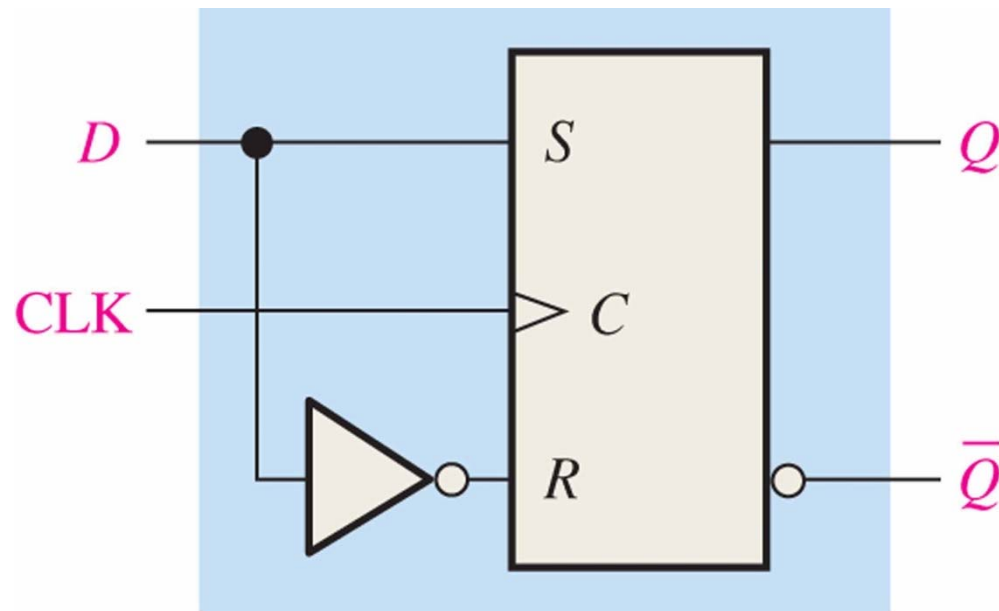
(b) A type of pulse transition detector

**Narrow
spike**





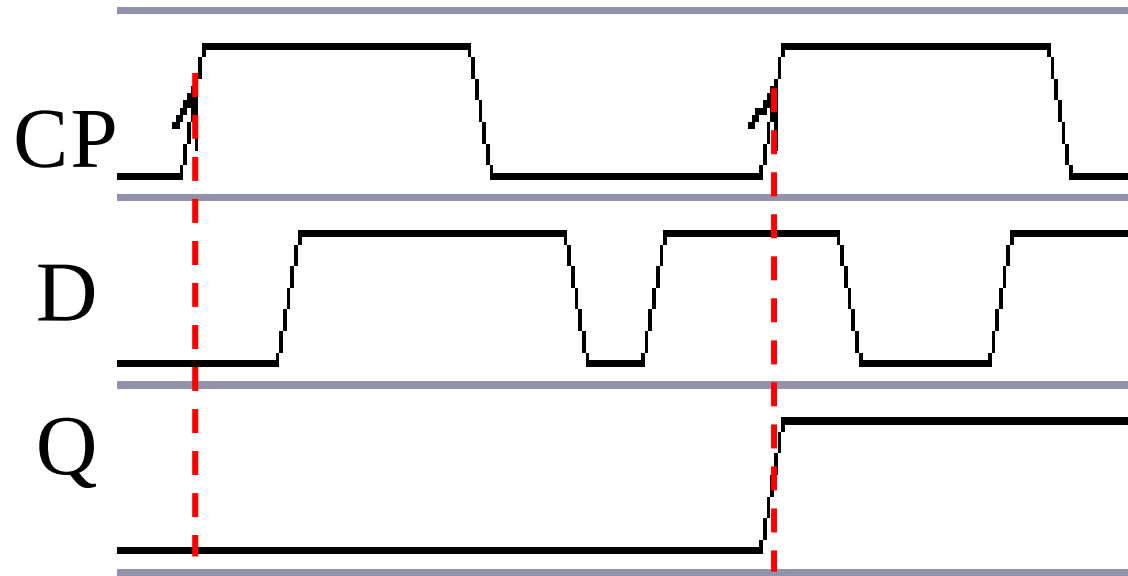
7.2.2 The Edge-Triggered D FF



Positive edge-triggered D flip-flop

The Q output of a D assumes the state of the D input on the triggering edge of the clock

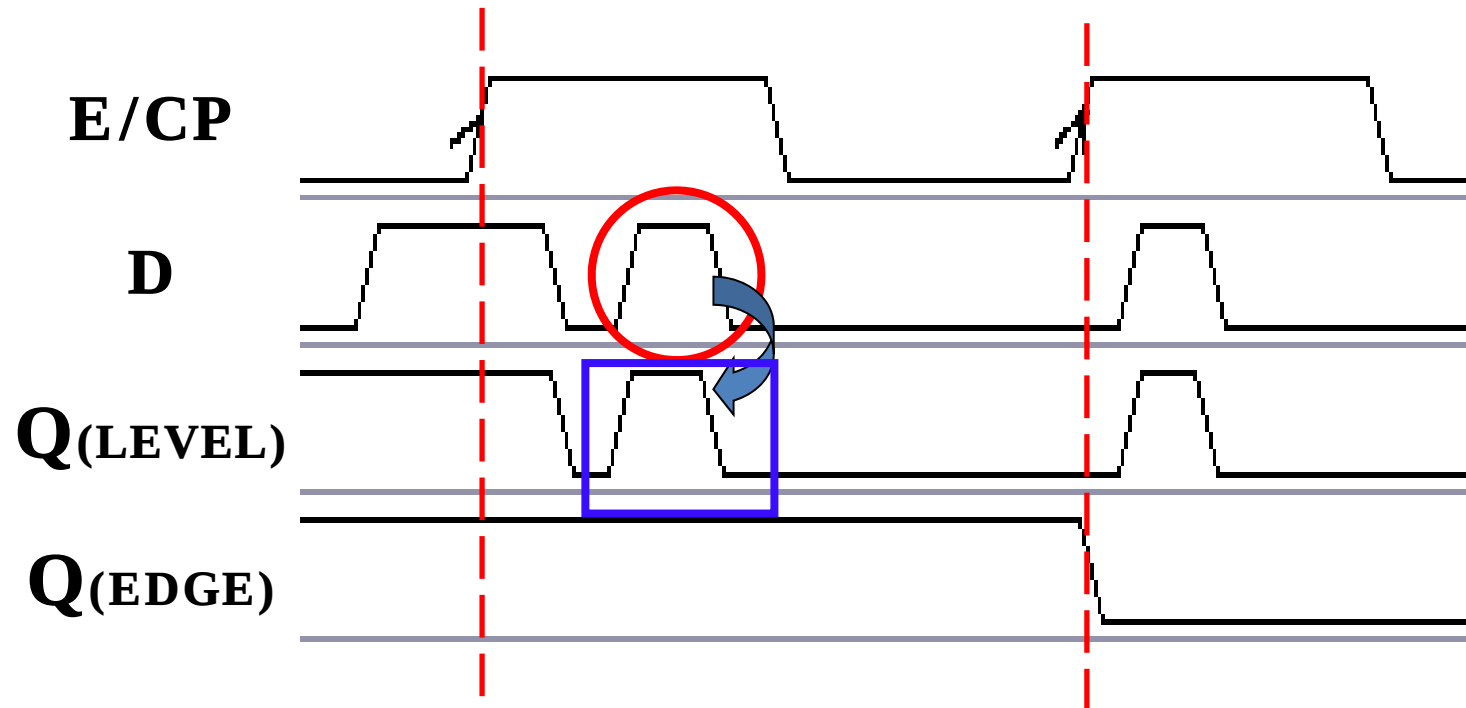
Timing diagram



Truth Table

CP	D	Q	$\overline{Q}$
$\uparrow$	D	D	$\overline{D}$

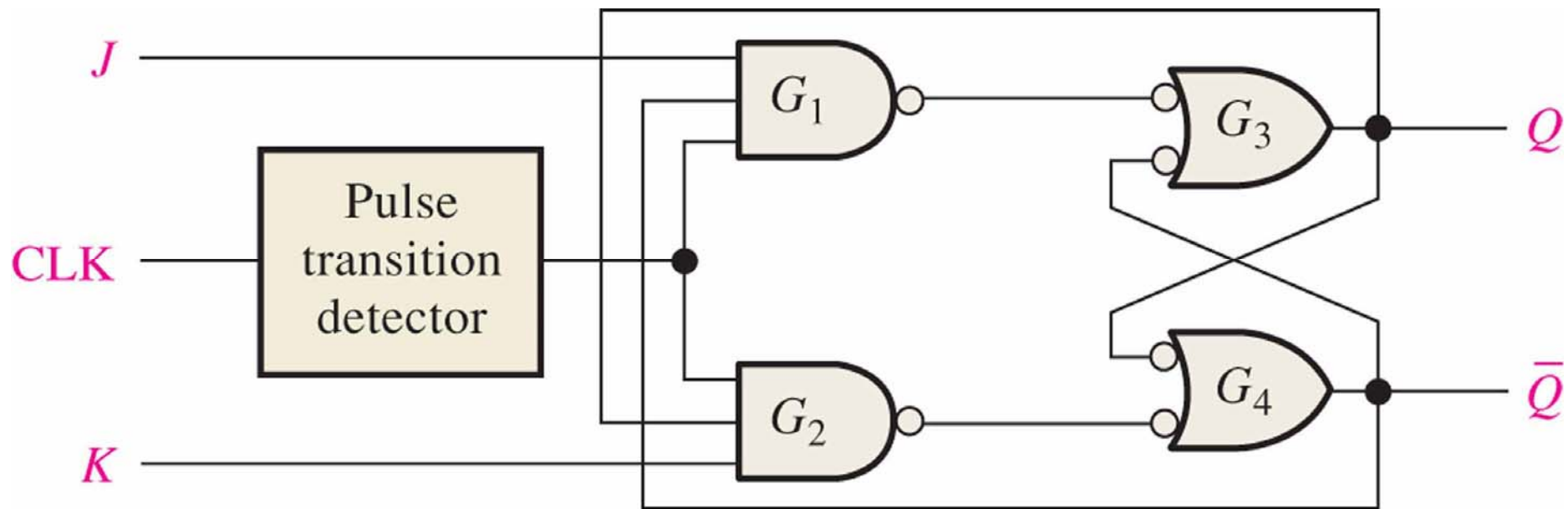
Comparison of edge-triggered and level-triggered



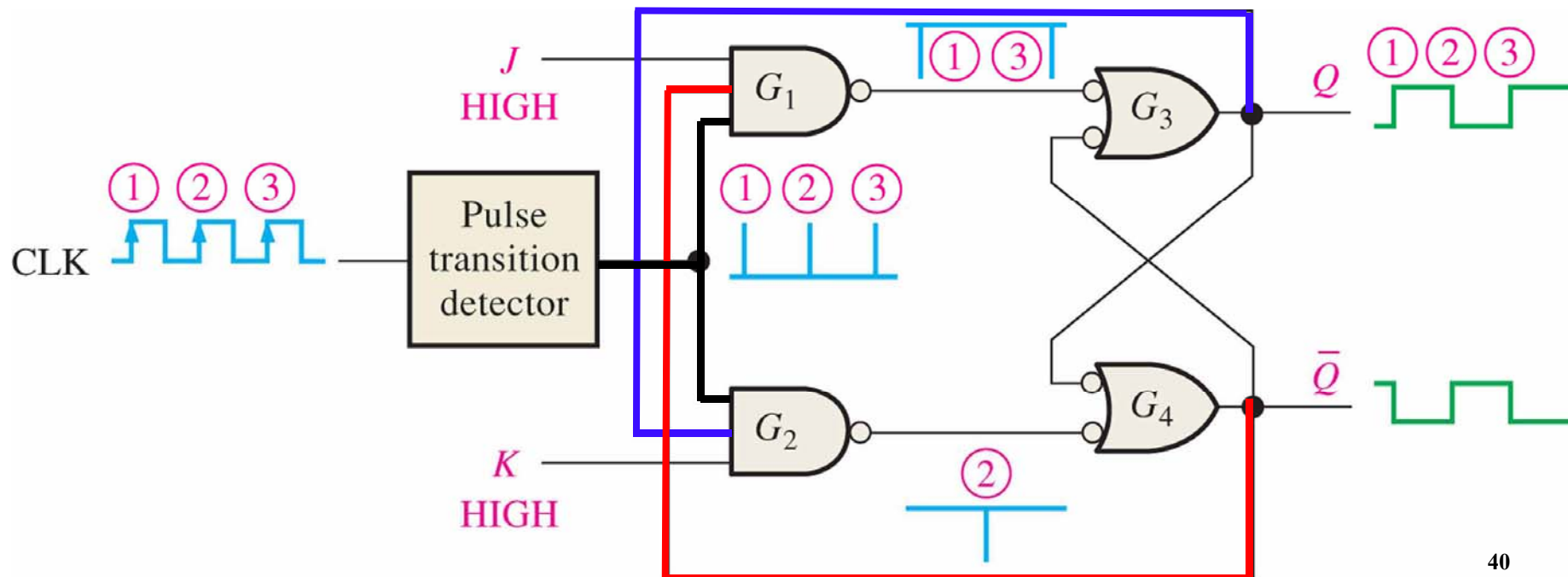
7.2.3 The Edge-Triggered J-K FF

****No invalid states.**

The two Set/Reset signals are labeled as J and K in honor of Jack Kilby.

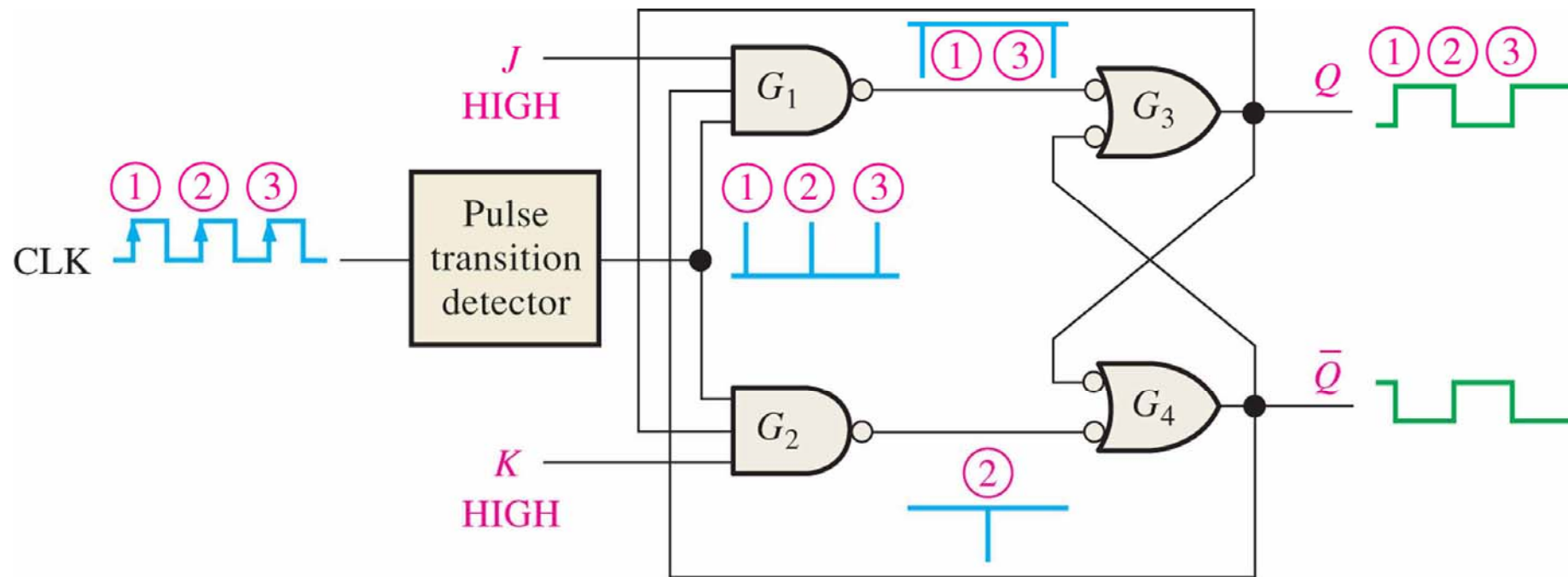


- ① IF $J=1, K=0, Q_0=0$; THEN G_1 enabled, $Q=1$ (SET)
- ② IF $J=0, K=1, Q_0=1$; THEN G_2 enabled, $Q=0$ (RESET)
- ③ IF $J=0, K=0$; THEN no change
- ④ IF $J=1, K=1$; THEN change to opposite state(*Toggle*)

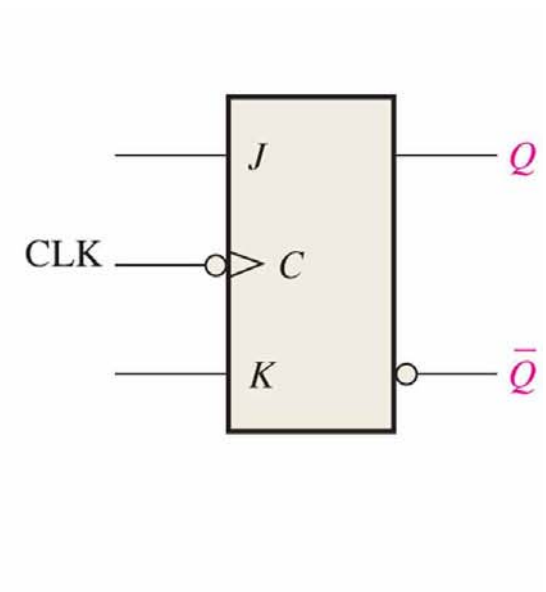
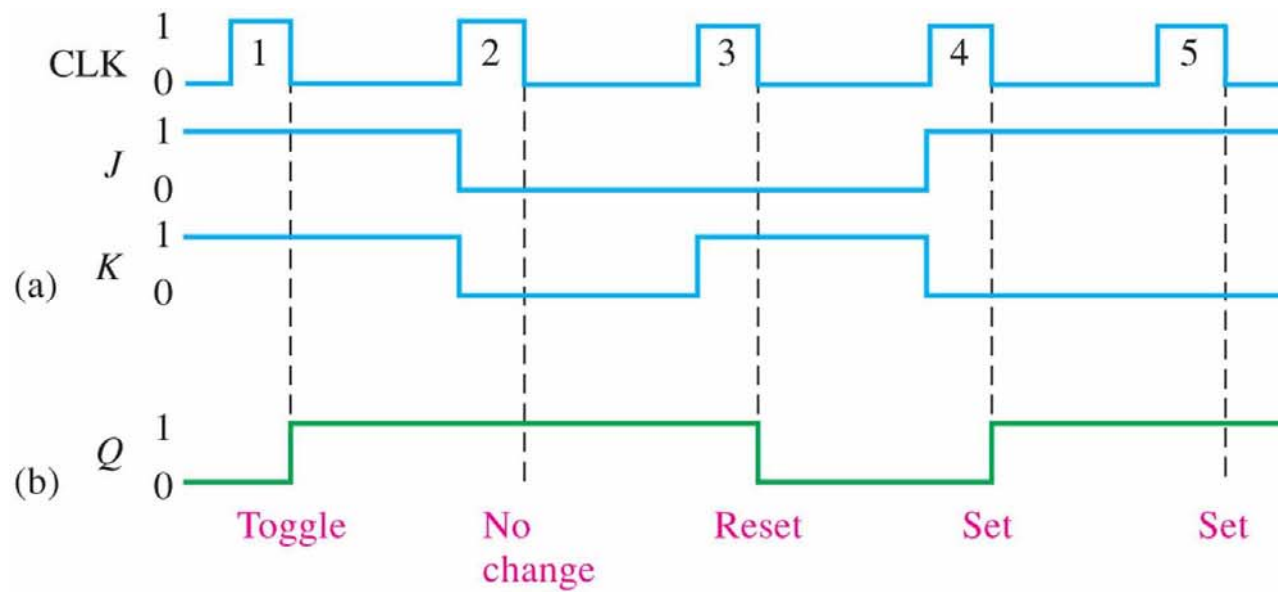


The Edge-Triggered Toggle FF

Toggle operation when $J=K=1$



Timing diagram



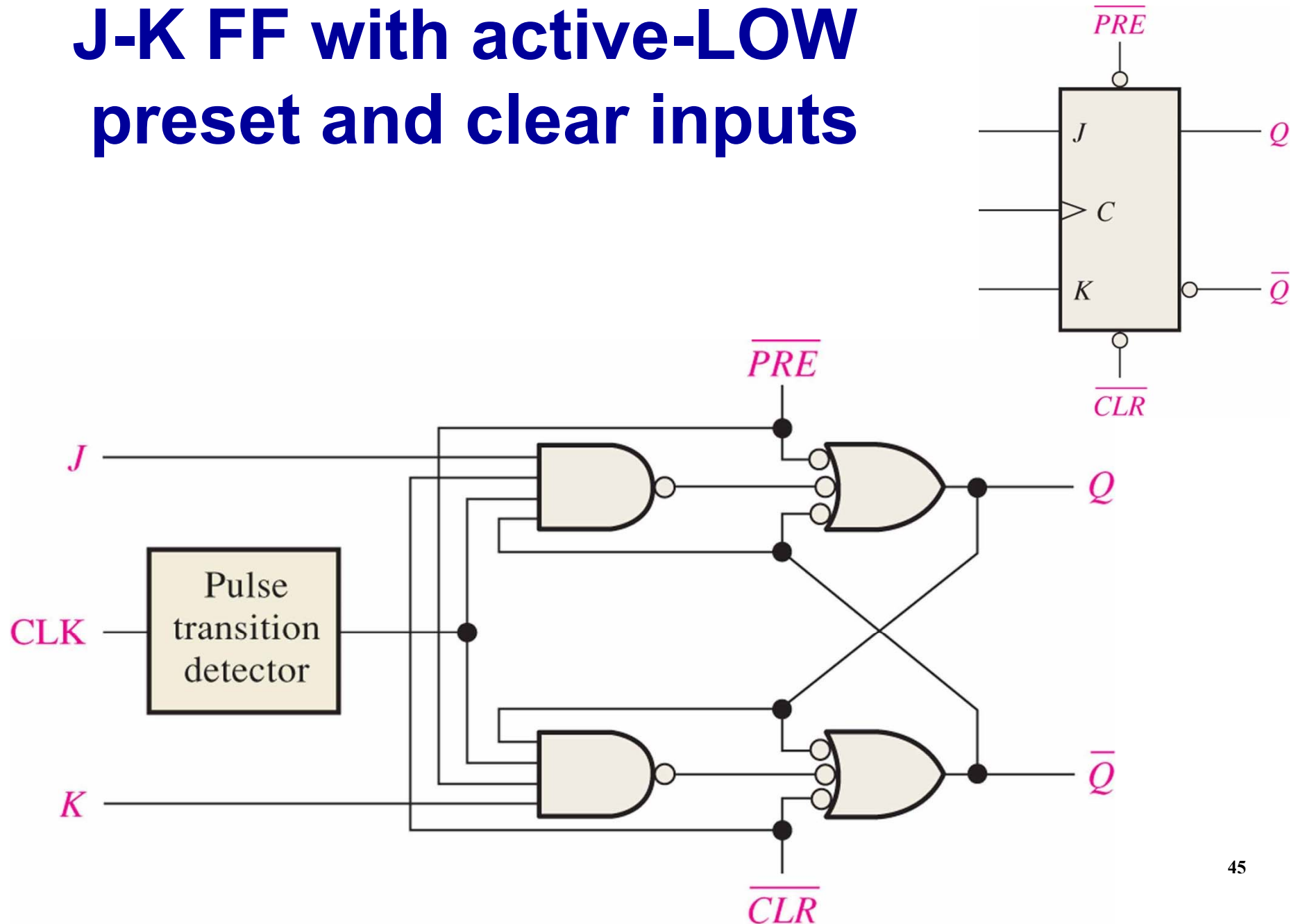
Truth Table

Inputs			Outputs		Comments
J	K	CLK	Q	$\bar{Q}$	
0	0	$\uparrow$	Q_0	$\bar{Q}_0$	No change
0	1	$\uparrow$	0	1	RESET
1	0	$\uparrow$	1	0	SET
1	1	$\uparrow$	$\bar{Q}_0$	Q_0	Toggle

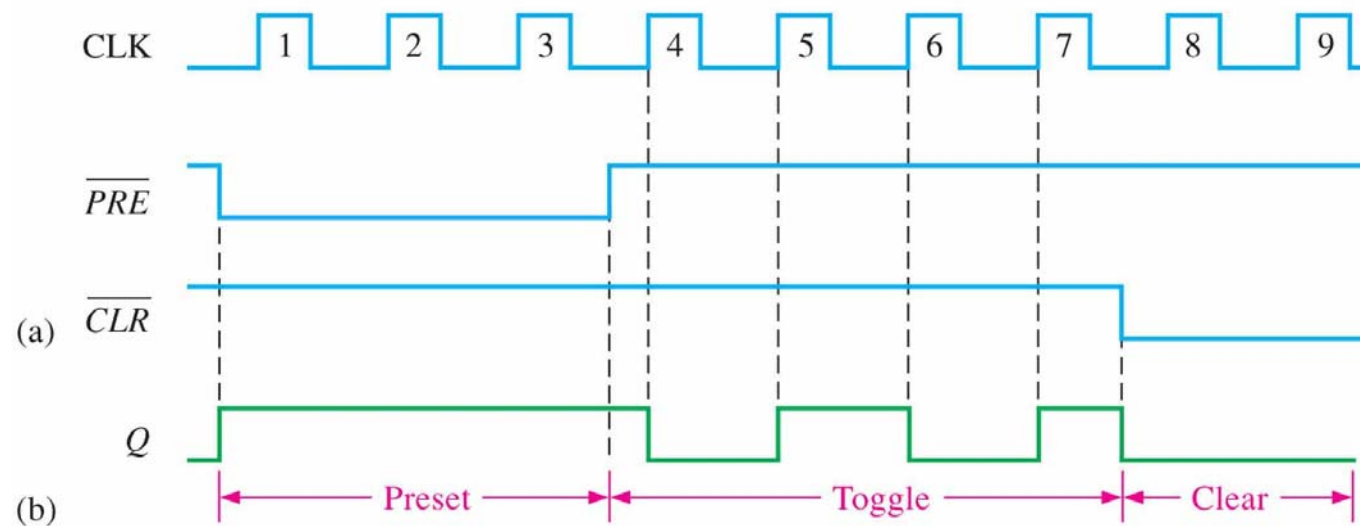
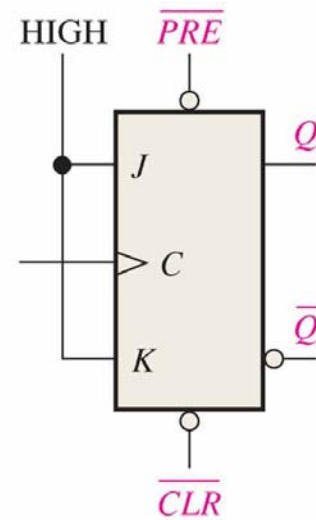
7.2.4. Asynchronous Preset and Clear Inputs (异步置位和复位)

- Asynchronous inputs are independent of the clock. Usually labeled *preset* (***PRE***) and *clear* (***CLR***), or *direct set* (***SD***) and *direct reset* (***RD***).
- An active level on the preset input will set the flip-flop, and an active level on the clear input will reset the flip-flop.

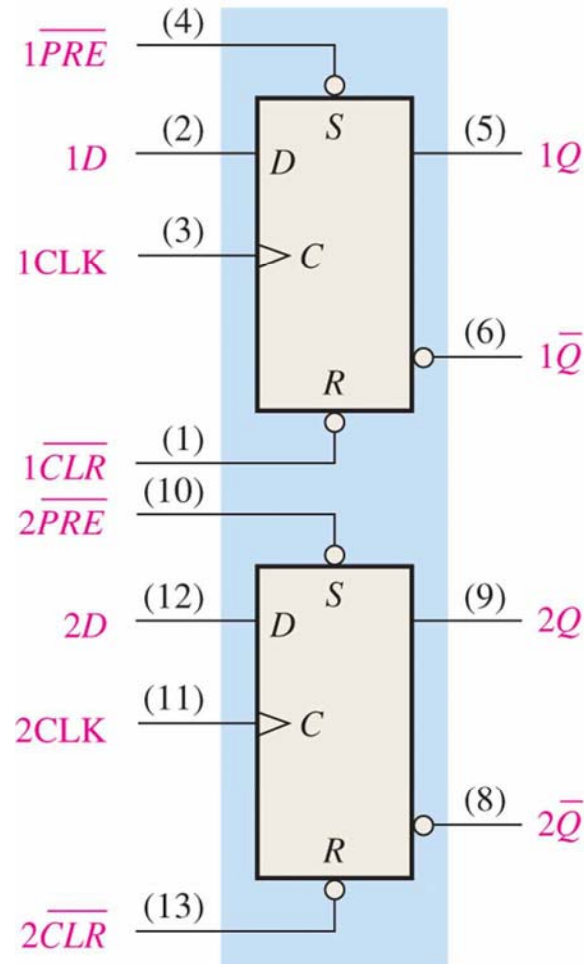
J-K FF with active-LOW preset and clear inputs



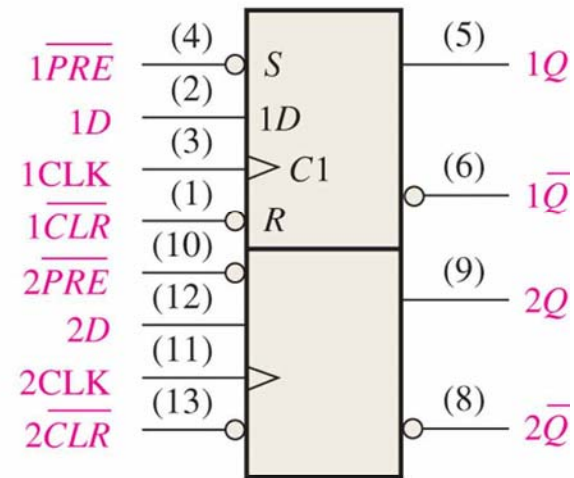
J-K FF with active-LOW preset and clear inputs



74AHC74 (dual D FF)



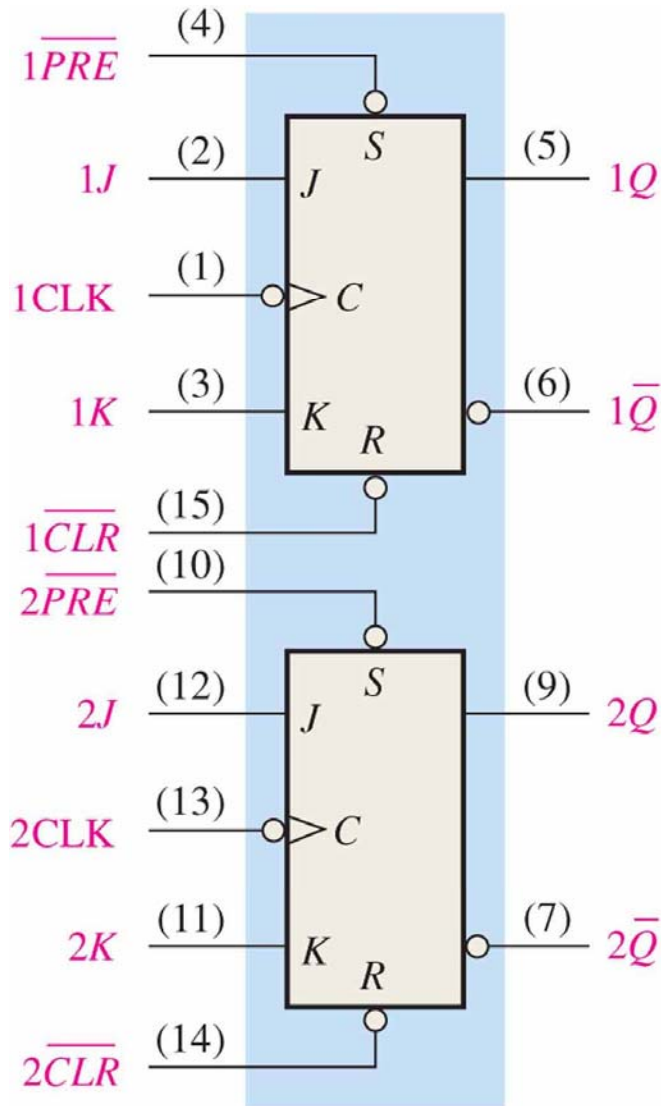
(a) Individual logic symbols



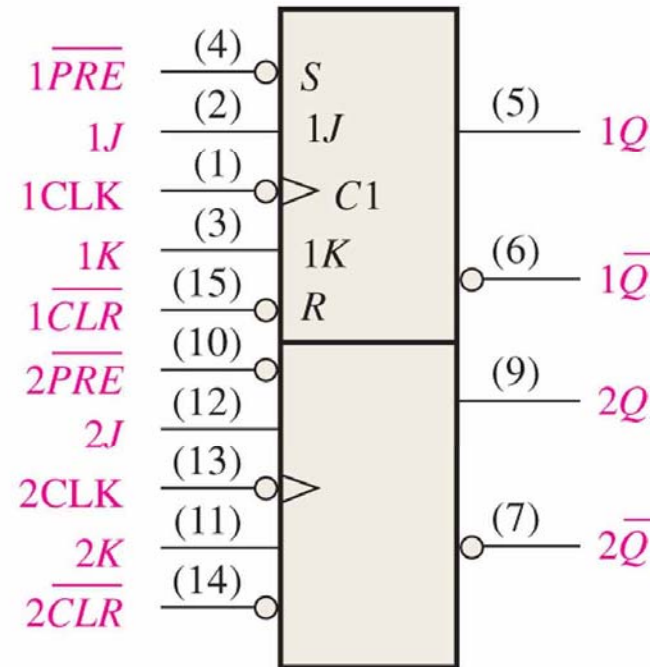
(b) Single block logic symbol

Note: The S and R inside the block indicate that $\overline{PRE}$ SETS and $\overline{CLR}$ RESETS.

74HC112 (dual JK FF)



(a) Individual logic symbols



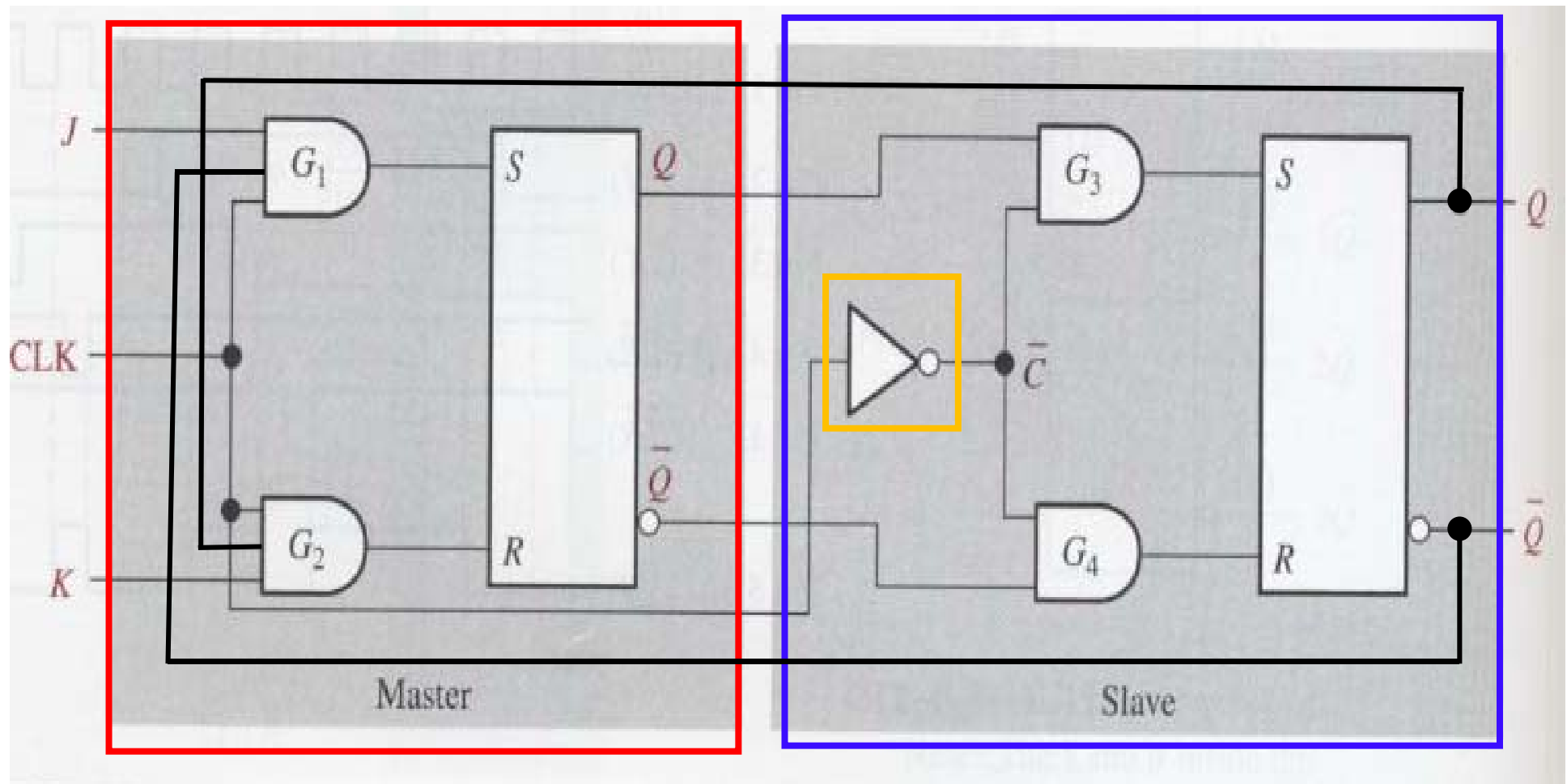
(b) Single block logic symbol

7.2.5 Master-slave Flip-Flops

主从触发器

- Composed of two section, the master section and slave section. The clock is inverted to the slave section.

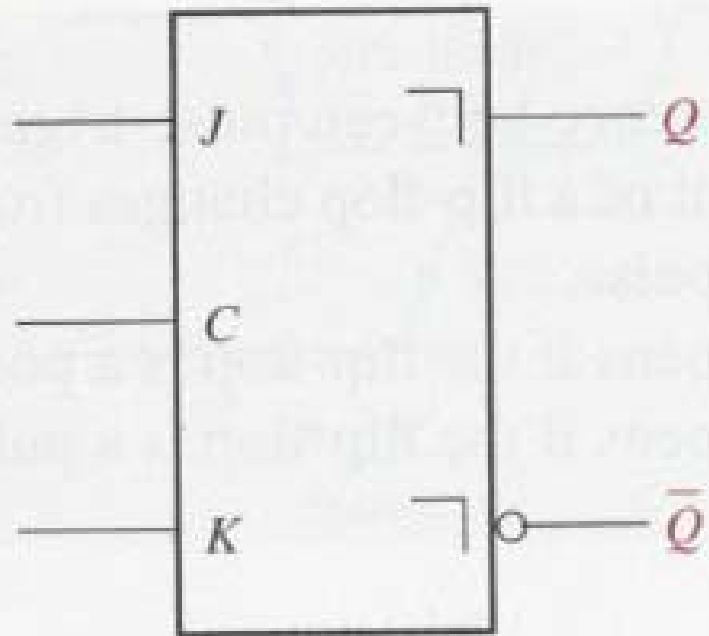
The Pulse-Triggered Master-Slave J-K Flip-Flop



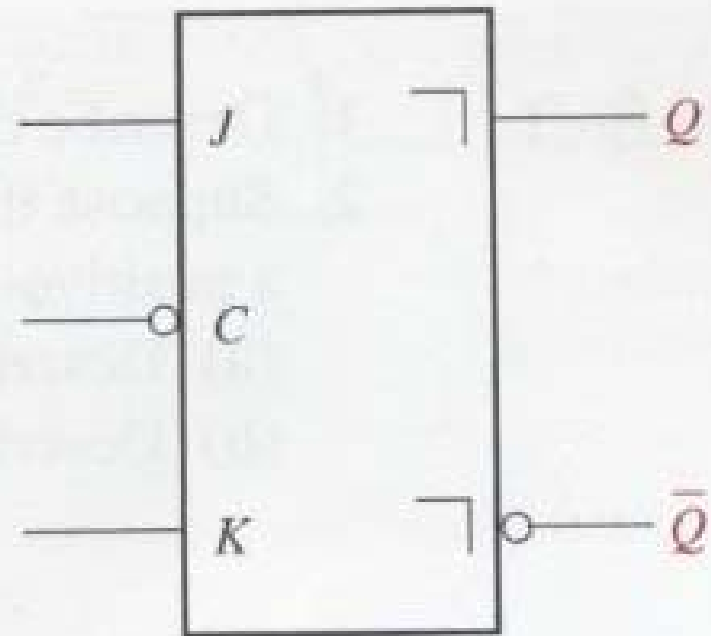
Truth Table

Inputs			Outputs		Comments
J	K	CLK	Q	$\bar{Q}$	
0	0		Q_0	$\bar{Q}_0$	No change
0	1		0	1	RESET
1	0		1	0	SET
1	1		$\bar{Q}_0$	Q_0	Toggle

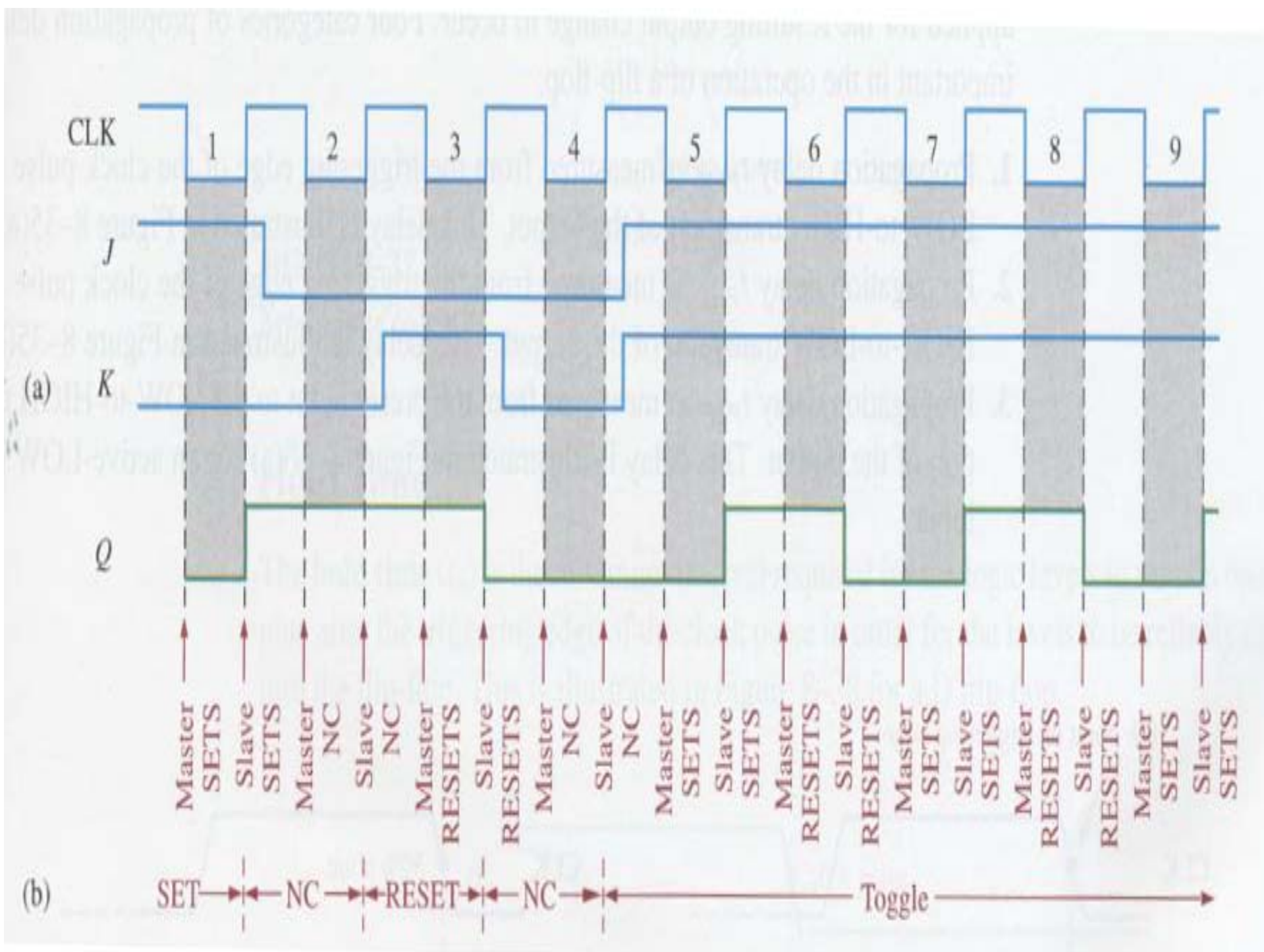
Logic Symbol



(a) Active-HIGH clock: Data are clocked in on positive-going edge of clock pulse and transferred to output on the following negative-going edge.



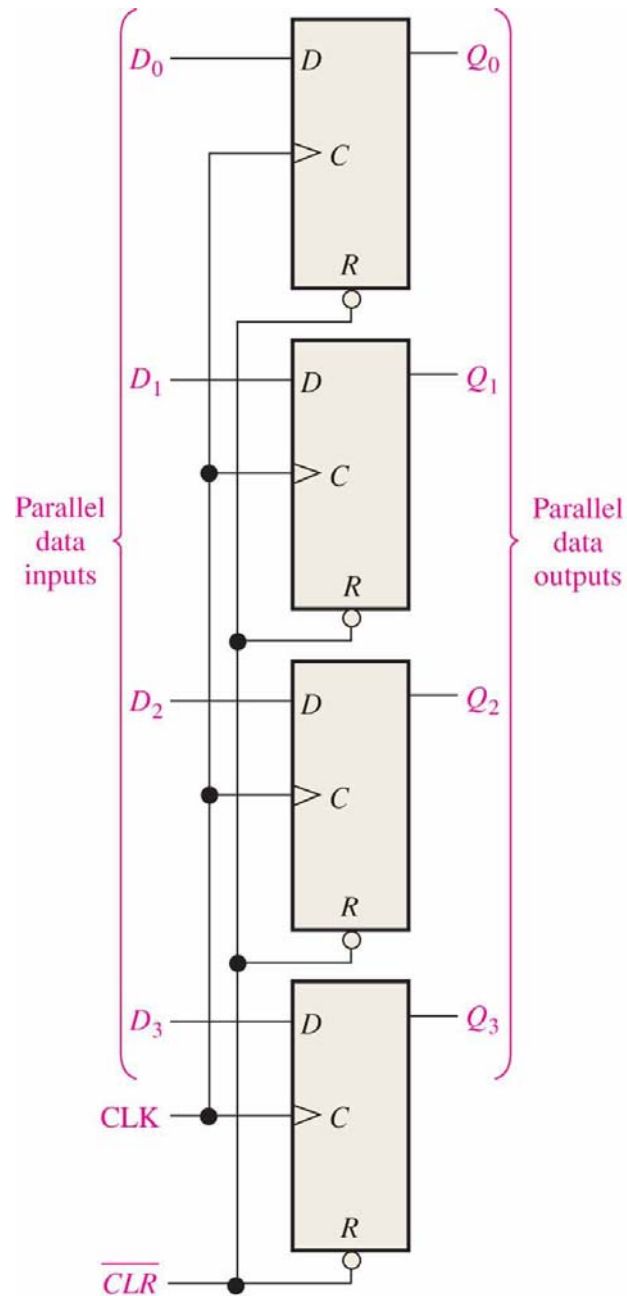
(b) Active-LOW clock: Data are clocked in on negative-going edge of clock pulse and transferred to output on the following positive-going edge.



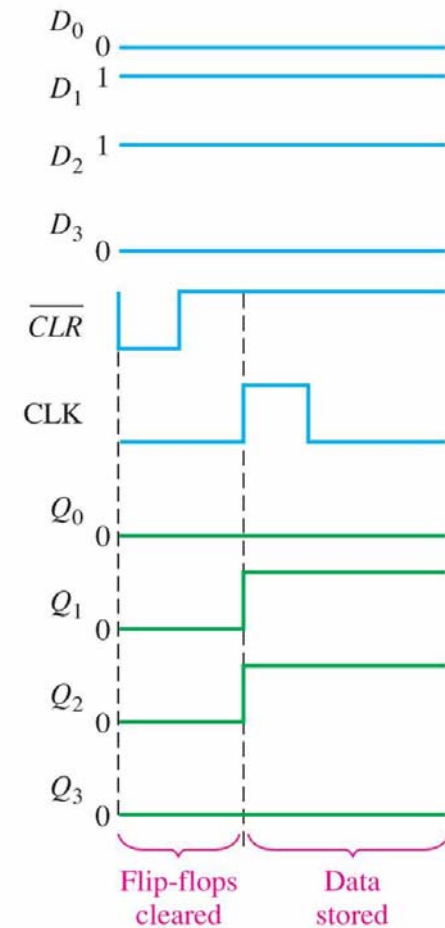
7.3 Flip-Flop Applications

- Parallel Data Storage (数据存储器)
- Frequency Division (分频器)
- Counter (计数器)

Application example: Basic register for parallel data storage

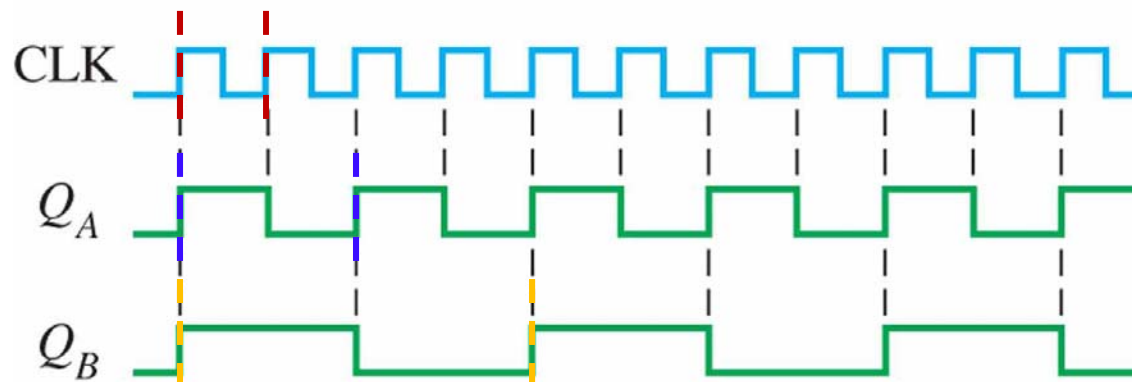
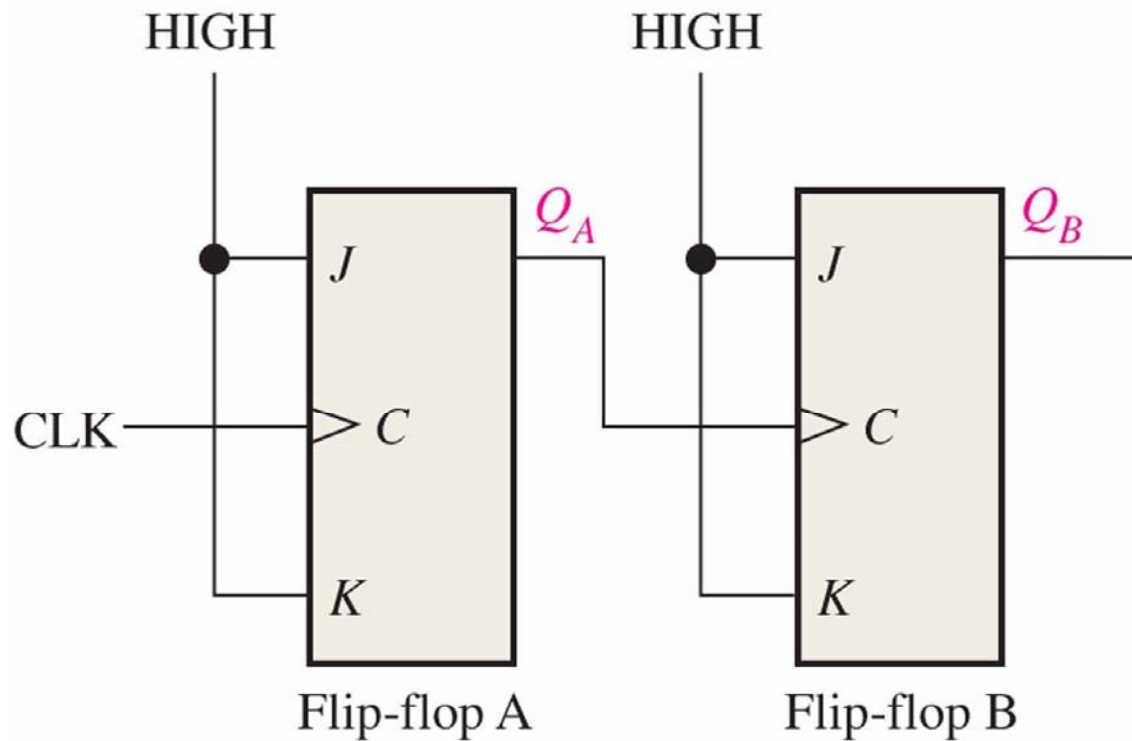


(a)

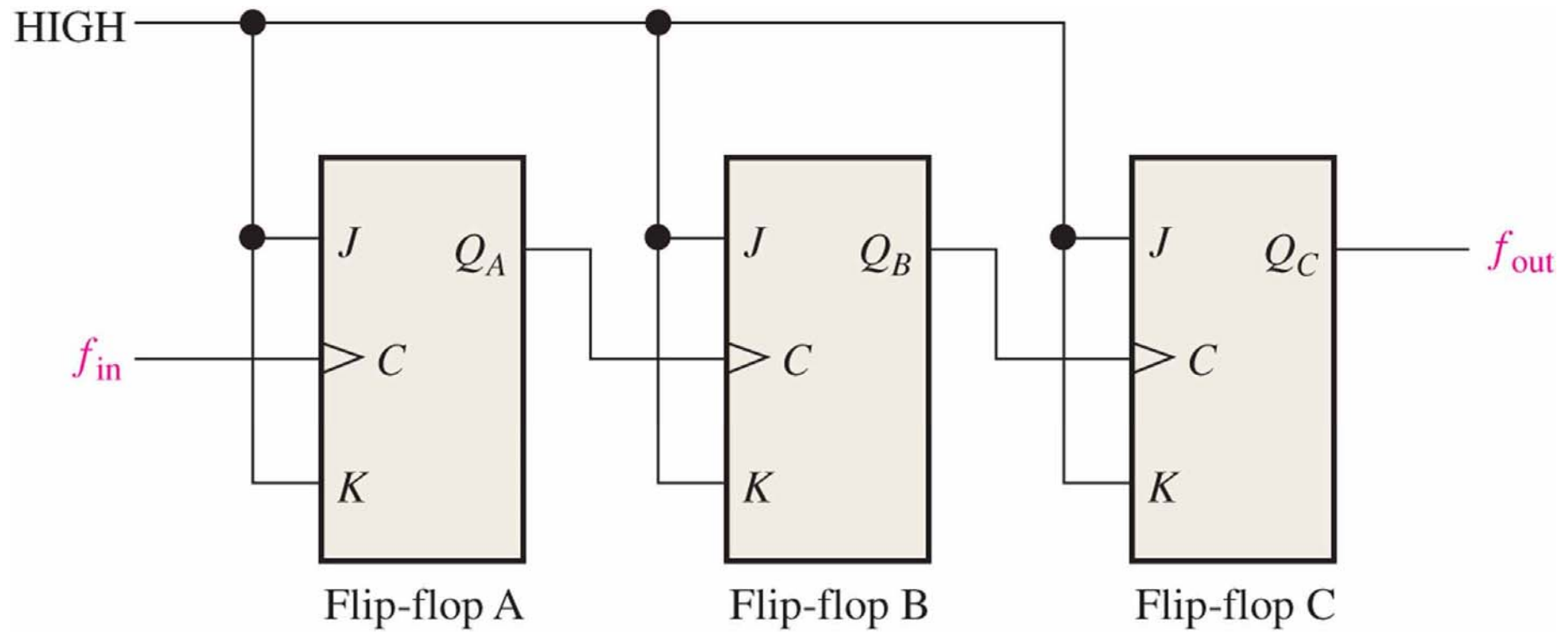


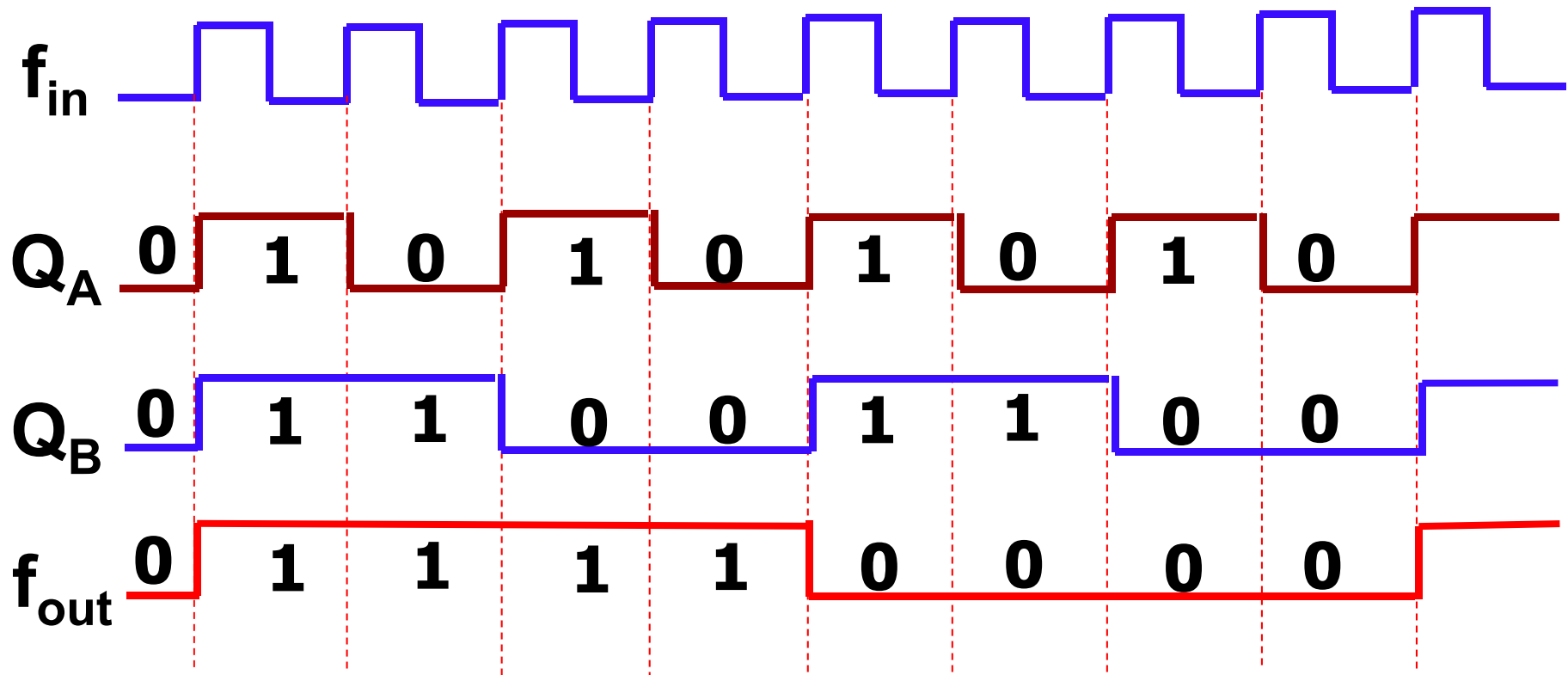
(b)

J-k FF as a divide-by-2 or 4 device



Exercise:





J-k FF as a counter

